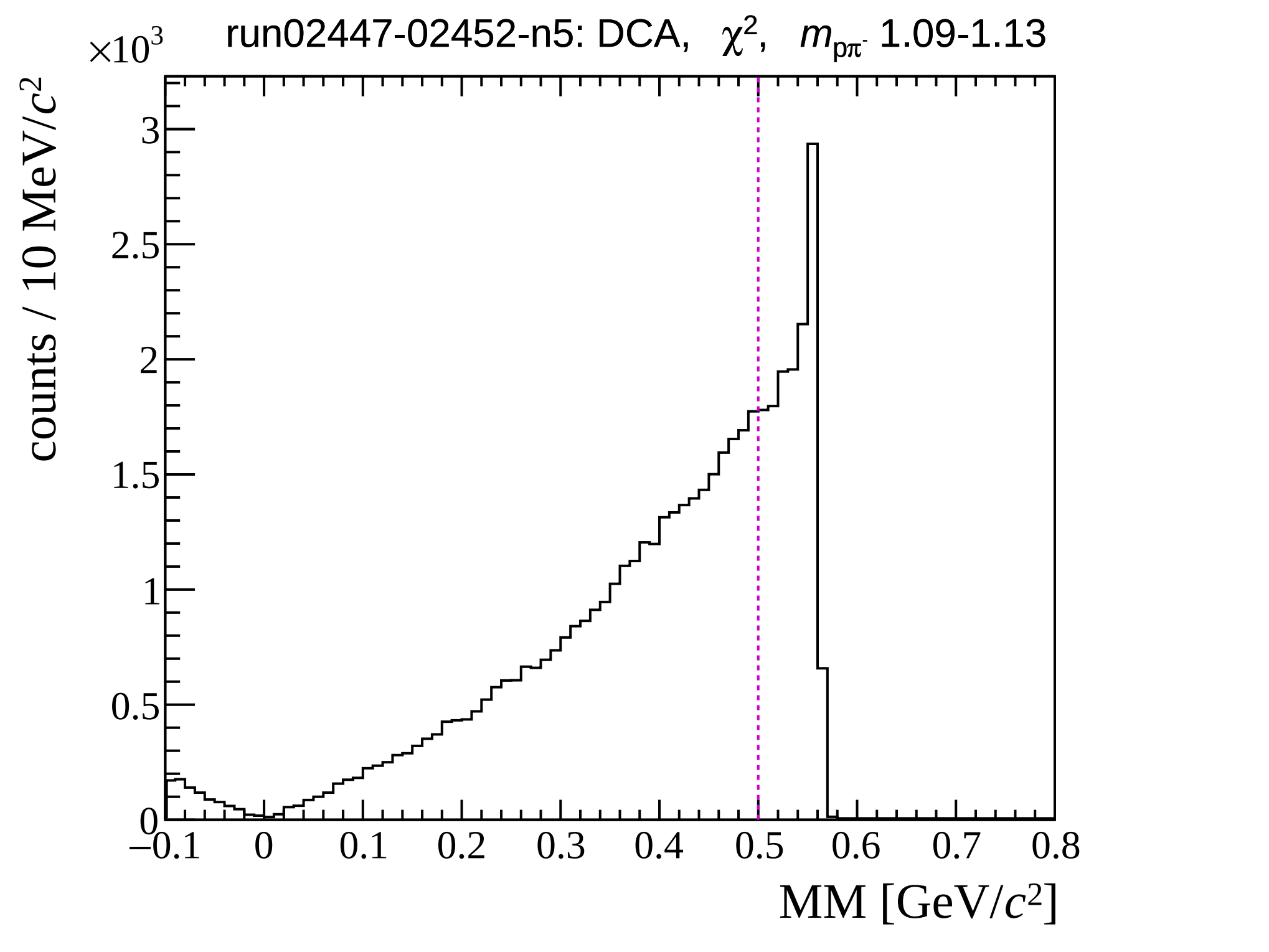
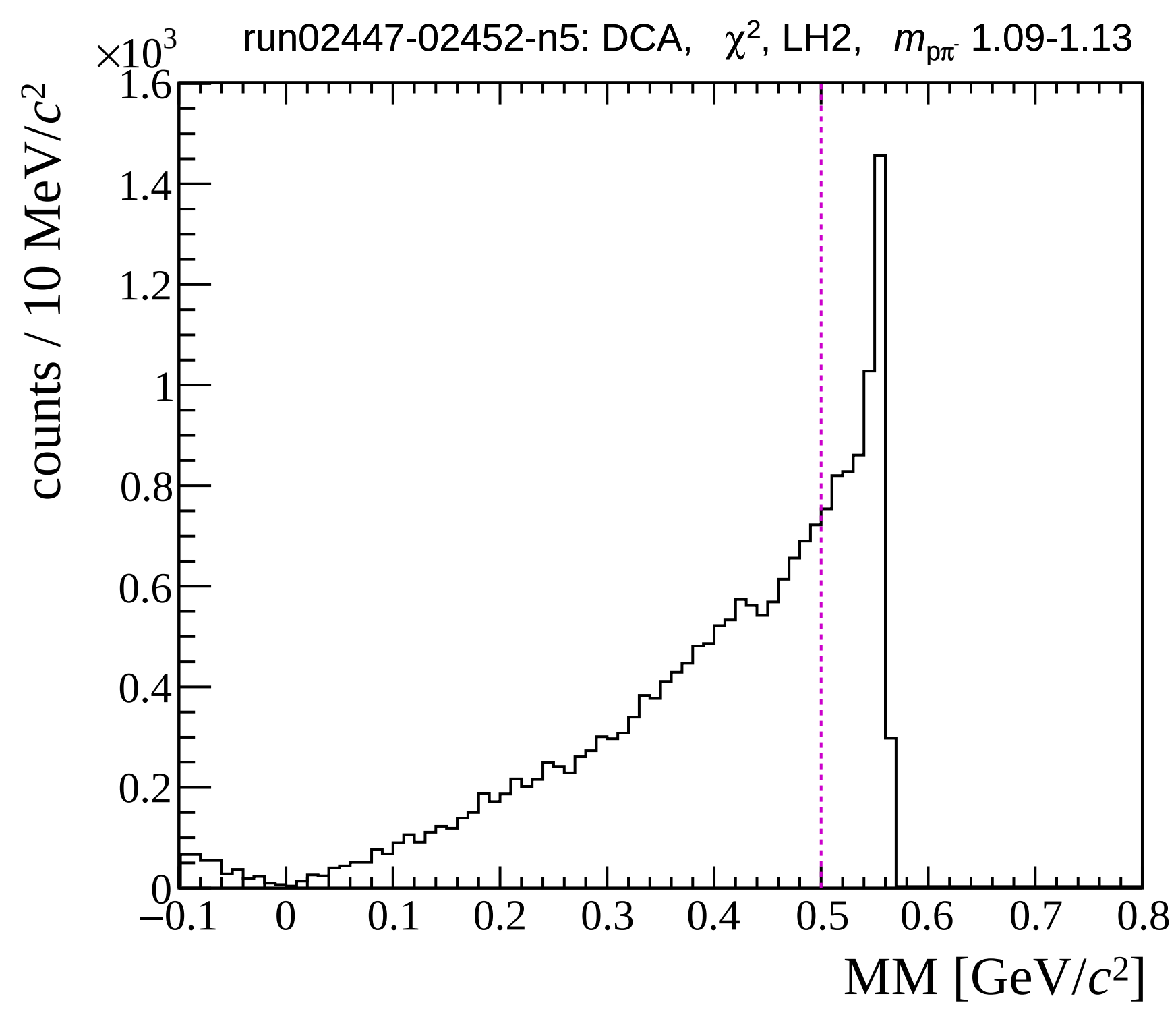
Data vs simulation, run02447-02452-n5
sim: result/yield-sim.root x TRIG-D x DAQ share of these runs (5) = 0.3230
     x eff_tpc_L 0.60, LH2 80 mm
sim: physics trigger, p and pi- of the Lambda each hit >= 5 TPC layers;
     ntrack: at most one other track hits >= 5 TPC layers (primary K- beam aside), as data;
     ntrack0: no other track; MM: true missing mass (K- p -> Lambda X) >= 0.5 GeV/c2;
     kin: true p_Lambda 0.4-0.6 and pi- at its first TPC hit < 0.25 GeV/c
data vertex ntrack: no good extra track within 30 mm of the production vertex;
     the simulation has no pile-up, so it is compared with the sim ntrack0

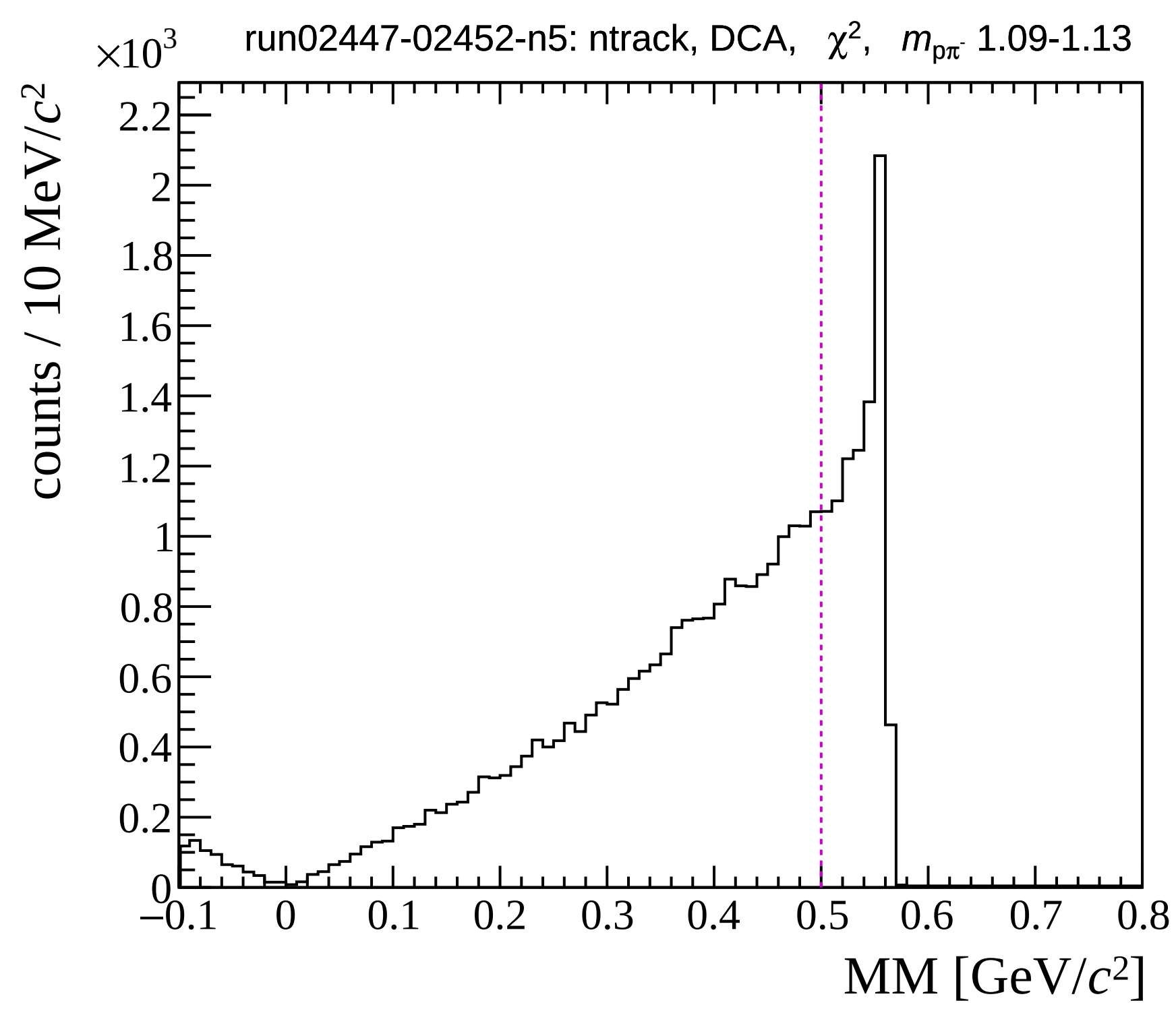
	all		ntrack		MM		ntrack, MM		
Simulation	all	LH2	all	LH2	all	LH2	all	LH2	
Lambda_eta	7895	4076	6264	3224	7895	4076	6264	3224	
Lambda_pi0	16287	7929	16265	7918	0	0	0	0	
Sigma0_pi0	12564	6160	12552	6156	0	0	0	0	
Sigmap_pim	9183	4479	9168	4474	1	0	0	0	(no Lambda)
Lambda_pip_pim	19273	9471	3383	1646	2948	1511	316	156	
Sigma0_pip_pim	3983	1968	817	400	1344	672	226	110	
Sigmap_pim_pi0	2898	1448	2895	1447	0	0	0	0	(no Lambda)
Lambda peak sum	60001	29604	39281	19344	12186	6259	6805	3489	
Sigma+ sum (no Lambda)	12081	5928	12063	5922	1	0	0	0	

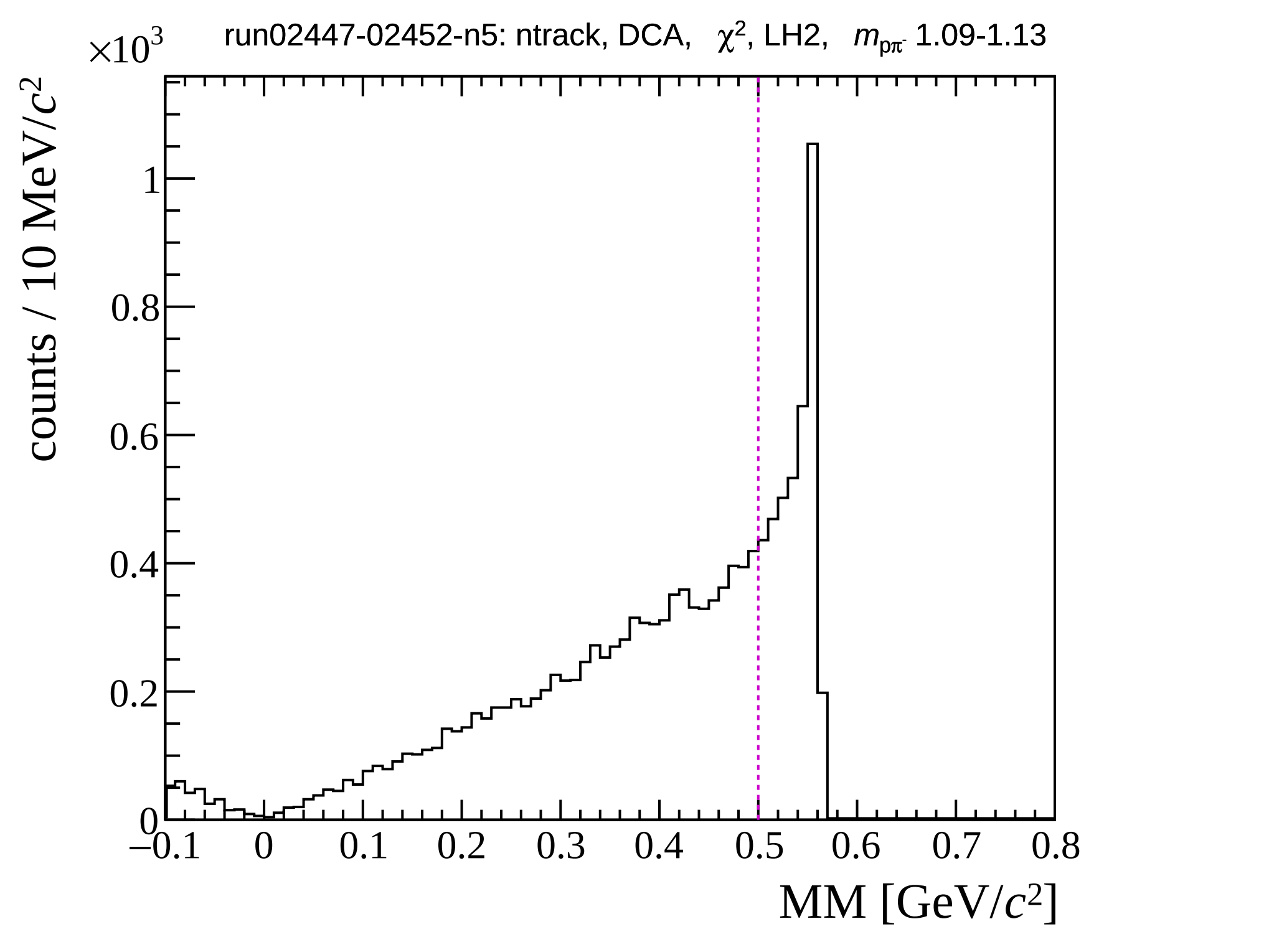
Lambda peak: ntrack0 33622 ntrack0_lh2 16573 ntrack0_mm 5528 ntrack0_mm_lh2 2848 kin_lh2 10652
Lambda peak: ntrack_kin_lh2 6228 ntrack0_kin_lh2 5186 mm_kin_lh2 5045 ntrack_mm_kin_lh2 3084 ntrack0_mm_kin_lh2 2577
Lambda peak: kin 21284 ntrack_kin 12409 ntrack0_kin 10310

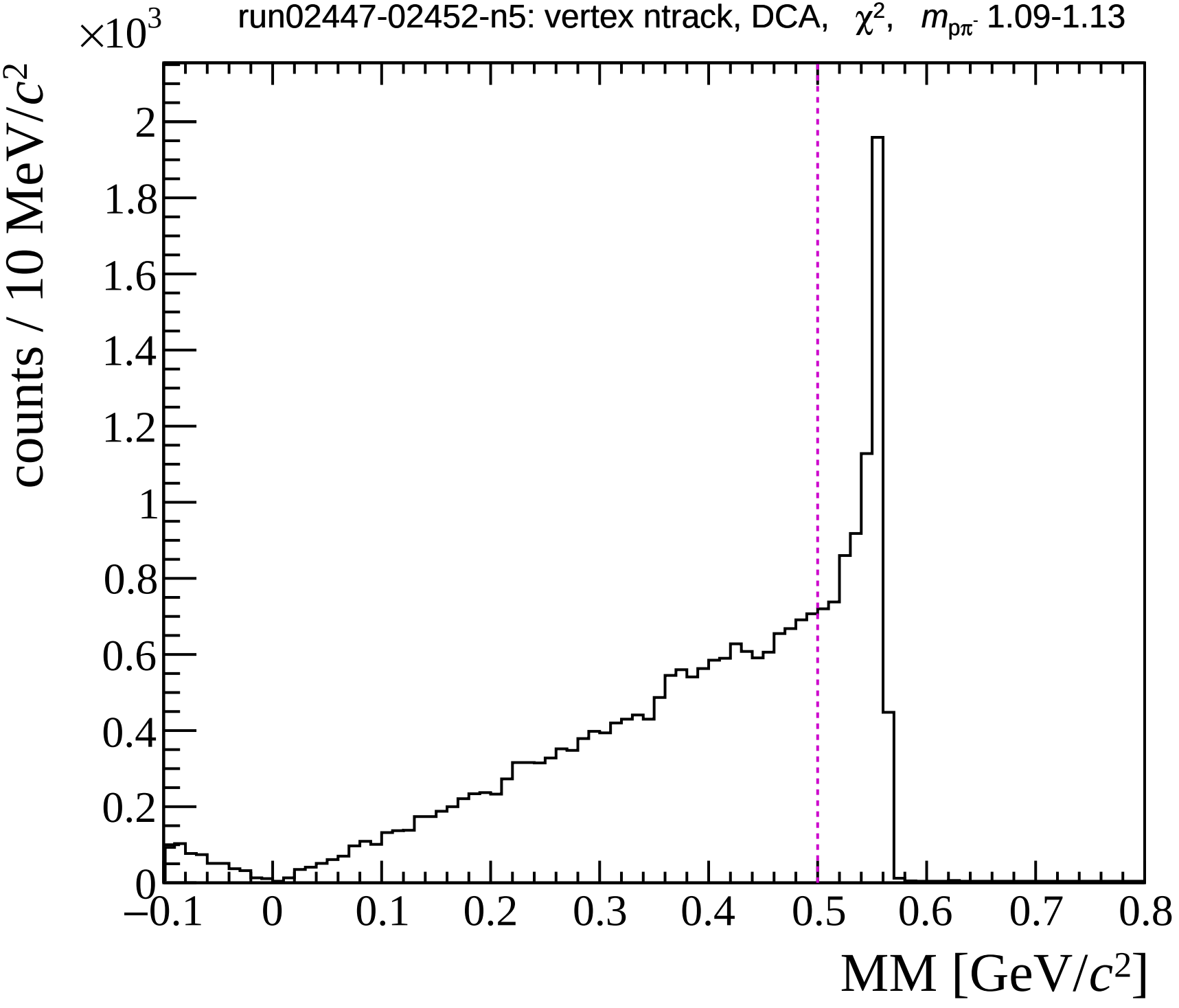
Data N_sig (fit)	N_sig	Lambda peak sim	data/sim
no cut	50828	60001 (tpc)	0.847
DCA, chi2	45581	60001 (tpc)	0.760
DCA, chi2, LH2	19276	29604 (tpc_lh2)	0.651
ntrack, DCA, chi2	31022	39281 (ntrack)	0.790
ntrack, DCA, chi2, LH2	13119	19344 (ntrack_lh2)	0.678
DCA, chi2, MM >= 0.5	10623	12186 (mm)	0.872
DCA, chi2, LH2, MM >= 0.5	5159	6259 (mm_lh2)	0.824
ntrack, DCA, chi2, MM >= 0.5	6931	6805 (ntrack_mm)	1.018
ntrack, DCA, chi2, LH2, MM >= 0.5	3198	3489 (ntrack_mm_lh2)	0.916
vertex ntrack, DCA, chi2	23942	33622 (ntrack0)	0.712
vertex ntrack, DCA, chi2, LH2	10516	16573 (ntrack0_lh2)	0.635
vertex ntrack, DCA, chi2, MM >= 0.5	5433	5528 (ntrack0_mm)	0.983
vertex ntrack, DCA, chi2, LH2, MM >= 0.5	2377	2848 (ntrack0_mm_lh2)	0.835
DCA, chi2, kin.	15396	21284 (kin)	0.723
ntrack, DCA, chi2, kin.	9626	12409 (ntrack_kin)	0.776
vertex ntrack, DCA, chi2, kin.	6999	10310 (ntrack0_kin)	0.679
DCA, chi2, LH2, kin.	6672	10652 (kin_lh2)	0.626
ntrack, DCA, chi2, LH2, kin.	3931	6228 (ntrack_kin_lh2)	0.631
vertex ntrack, DCA, chi2, LH2, kin.	3337	5186 (ntrack0_kin_lh2)	0.643
DCA, chi2, LH2, kin., MM >= 0.5	3273	5045 (mm_kin_lh2)	0.649
ntrack, DCA, chi2, LH2, kin., MM >= 0.5	1960	3084 (ntrack_mm_kin_lh2)	0.635
vertex ntrack, DCA, chi2, LH2, kin., MM >= 0.5	1477	2577 (ntrack0_mm_kin_lh2)	0.573

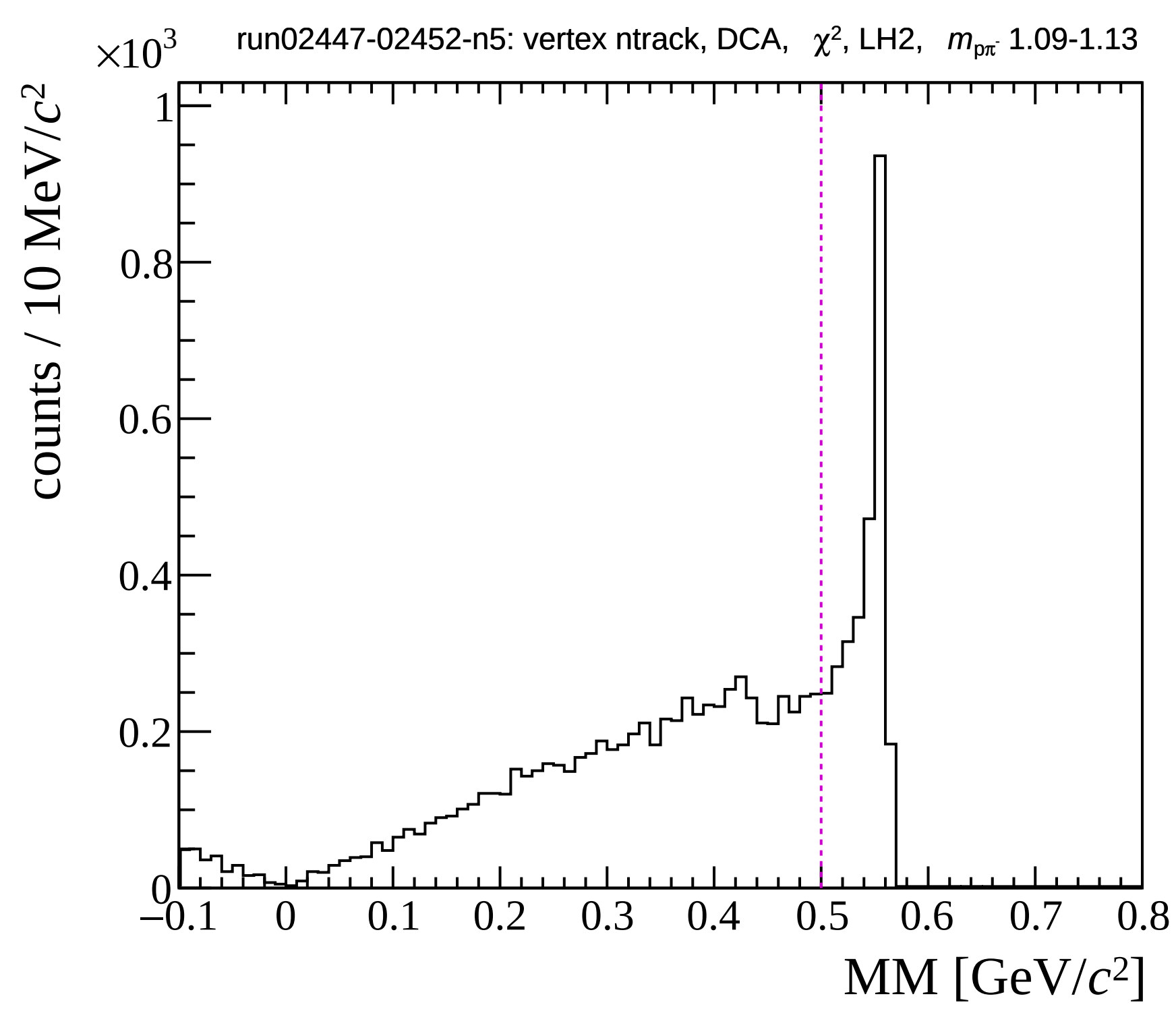

run02447-02452-n5: DCA, χ^2, $m_{p\pi^-}$ 1.09-1.13
×10^3
counts / 10 MeV/c^2
3
2.5
2
1.5
1
0.5
0
−0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c^2]

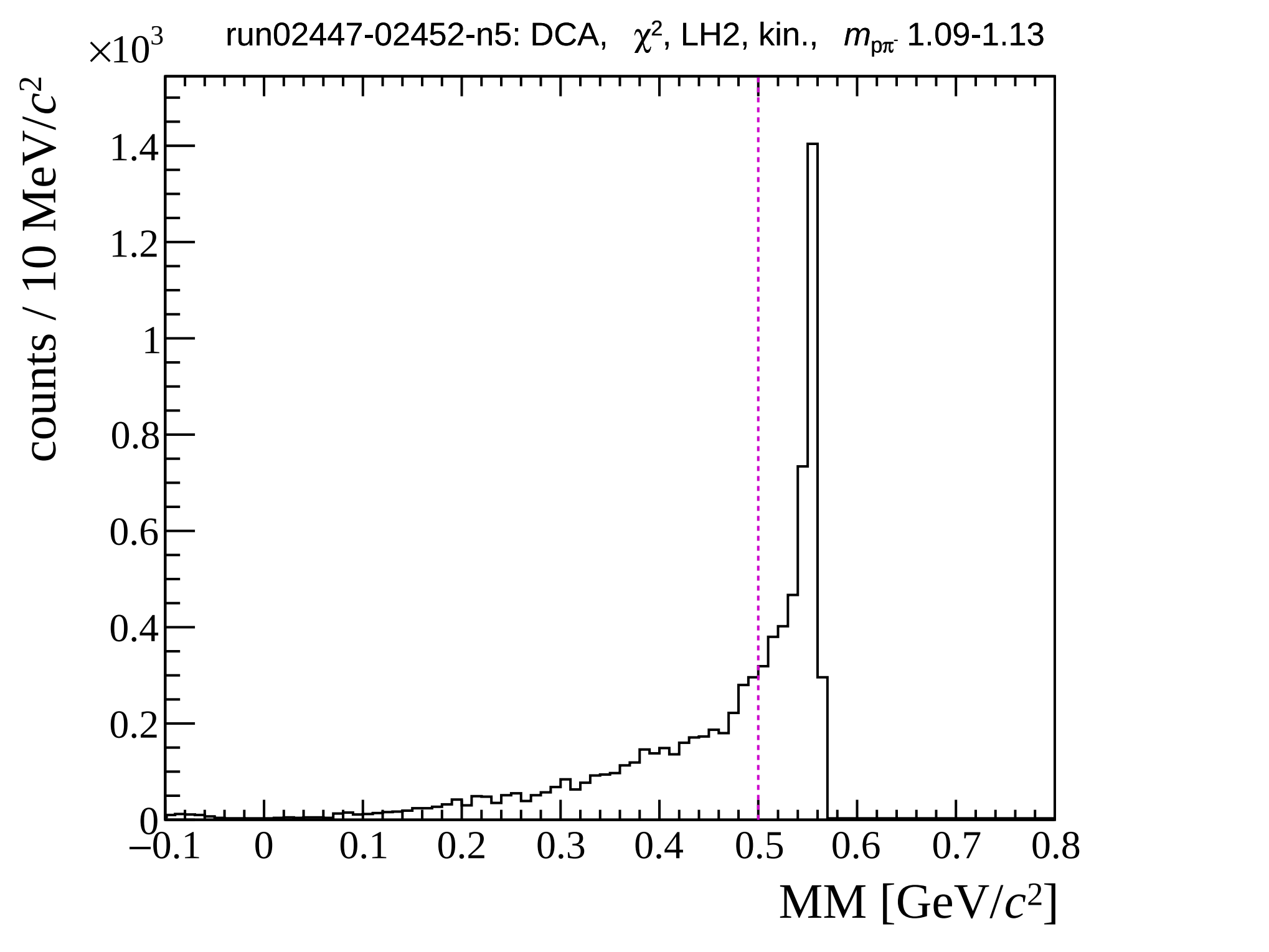

run02447-02452-n5: DCA, χ^2, LH2, $m_{p\pi^-}$ 1.09-1.13
$\times 10^3$
counts / 10 MeV/c^2
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
−0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c^2]

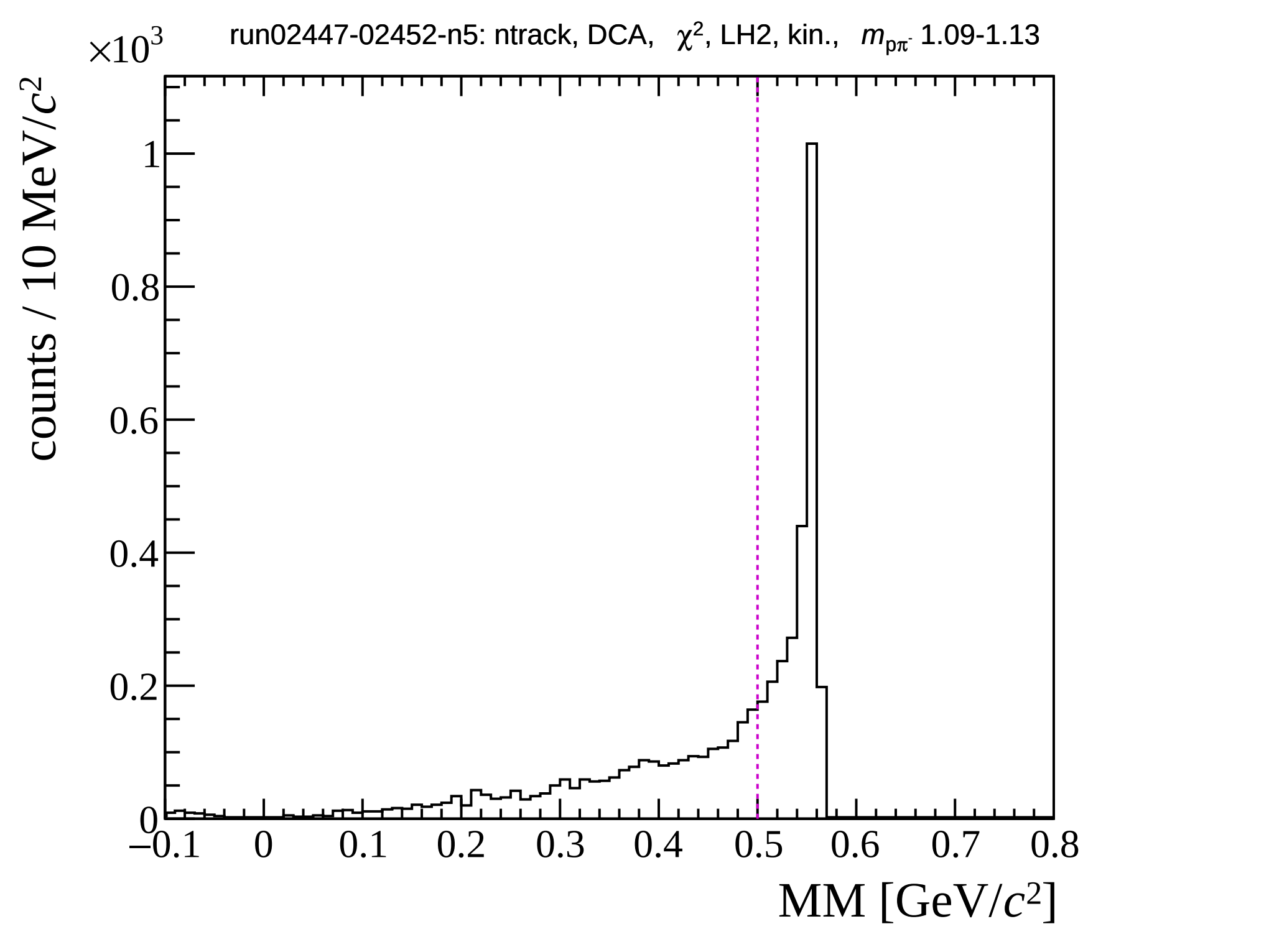

run02447-02452-n5: ntrack, DCA, χ^2, $m_{p\pi^-}$ 1.09-1.13
$\times 10^3$
counts / 10 MeV/c^2
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
−0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c^2]

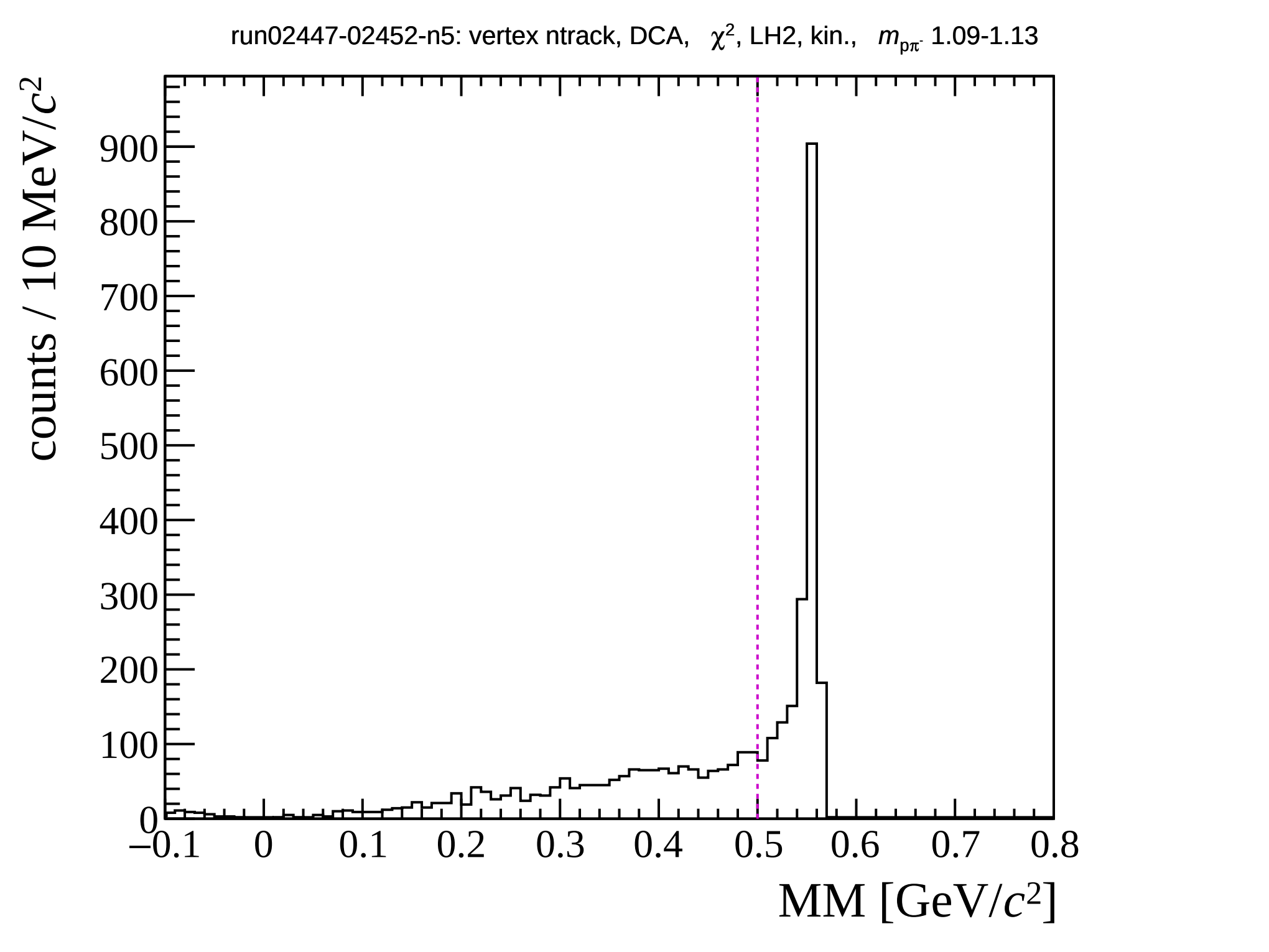

run02447-02452-n5: ntrack, DCA, χ^2, LH2, $m_{p\pi^-}$ 1.09-1.13
$\times 10^3$
counts / 10 MeV/c^2
1
0.8
0.6
0.4
0.2
0
−0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c^2]

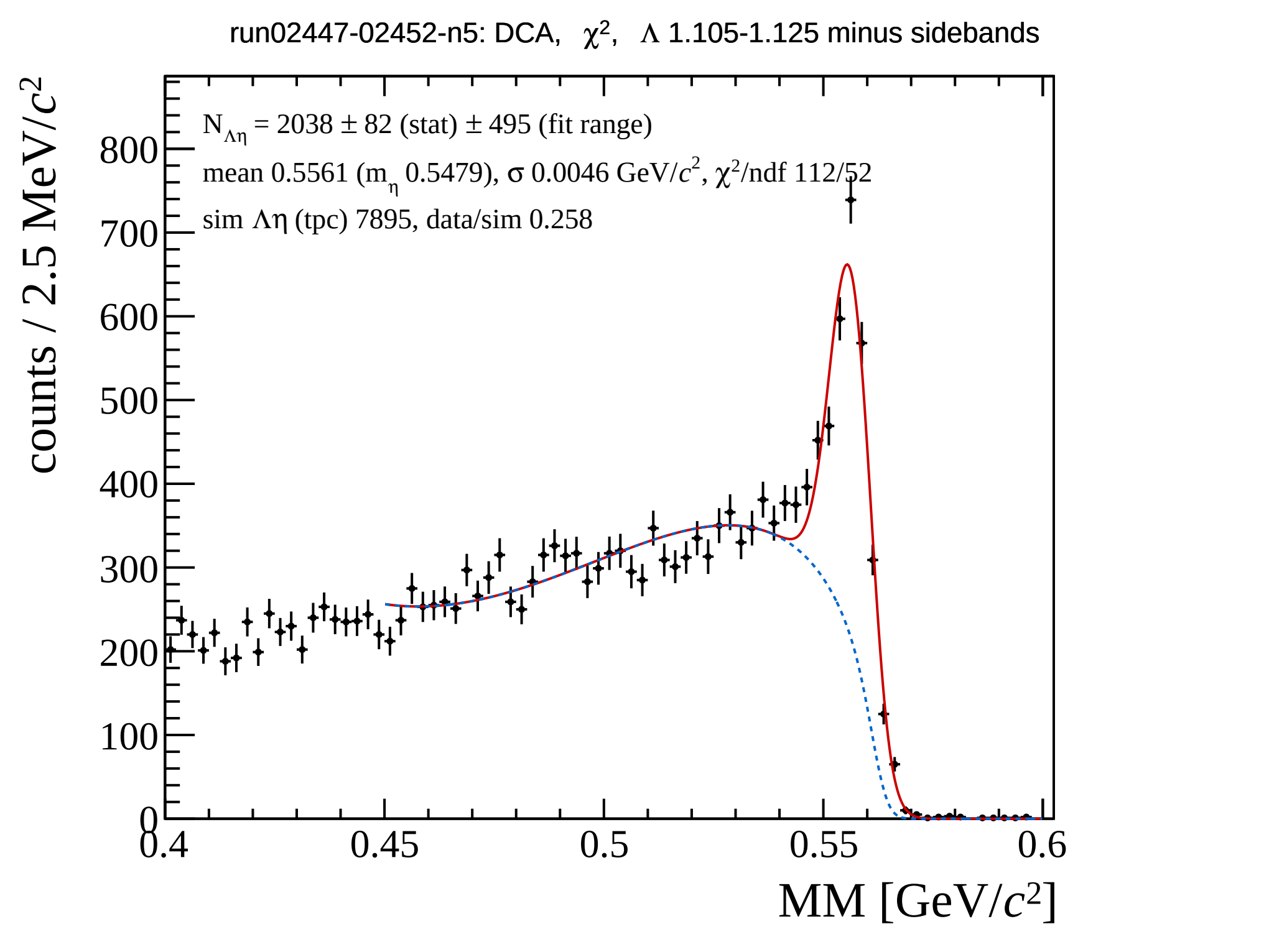

run02447-02452-n5: vertex ntrack, DCA, χ^2, $m_{p\pi^-}$ 1.09-1.13
$\times 10^3$
counts / 10 MeV/c^2
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
–0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c^2]

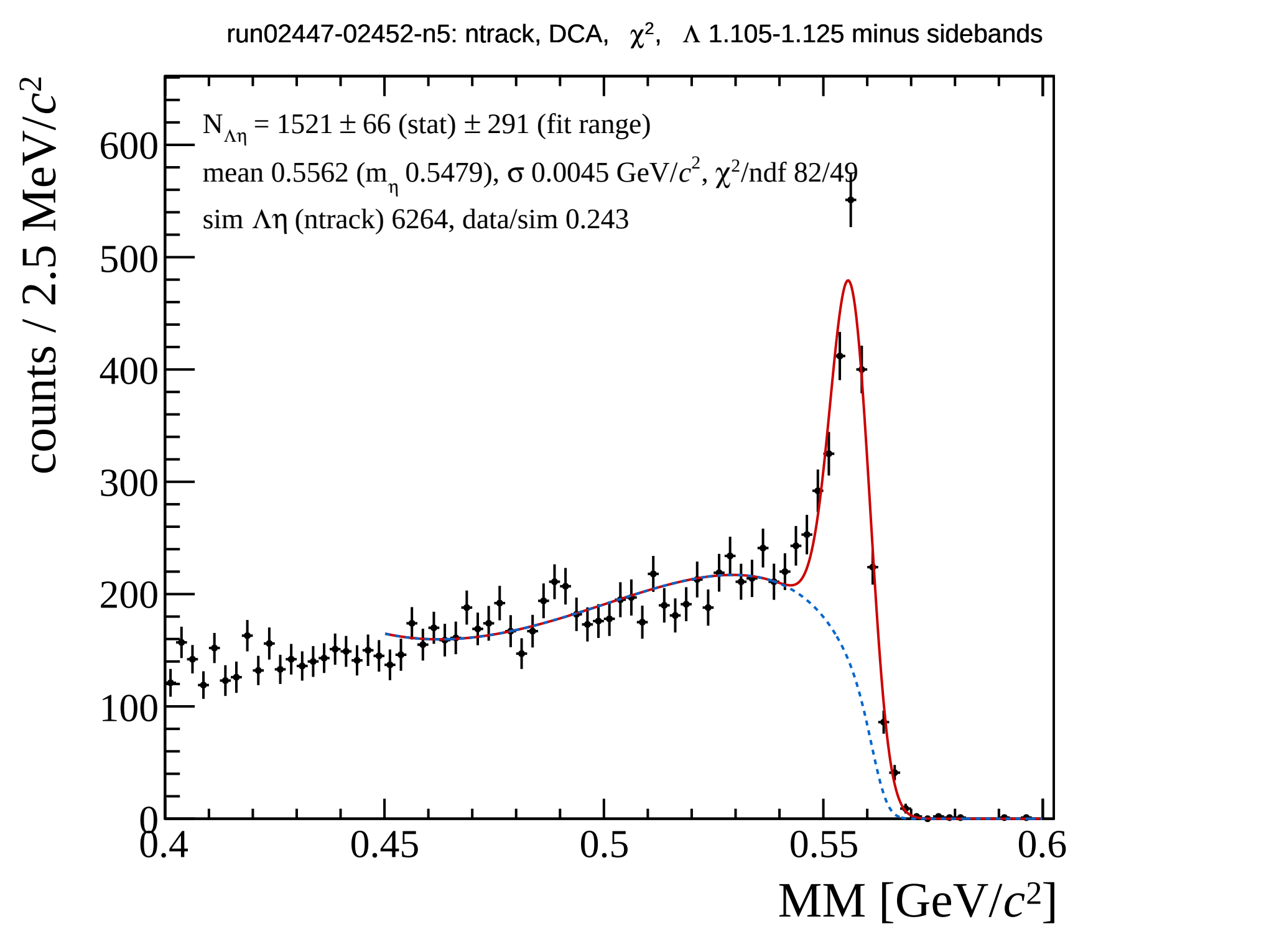

run02447-02452-n5: vertex ntrack, DCA, χ², LH2, m pπ- 1.09-1.13
×10³
counts / 10 MeV/c²
1
0.8
0.6
0.4
0.2
0
−0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c²]

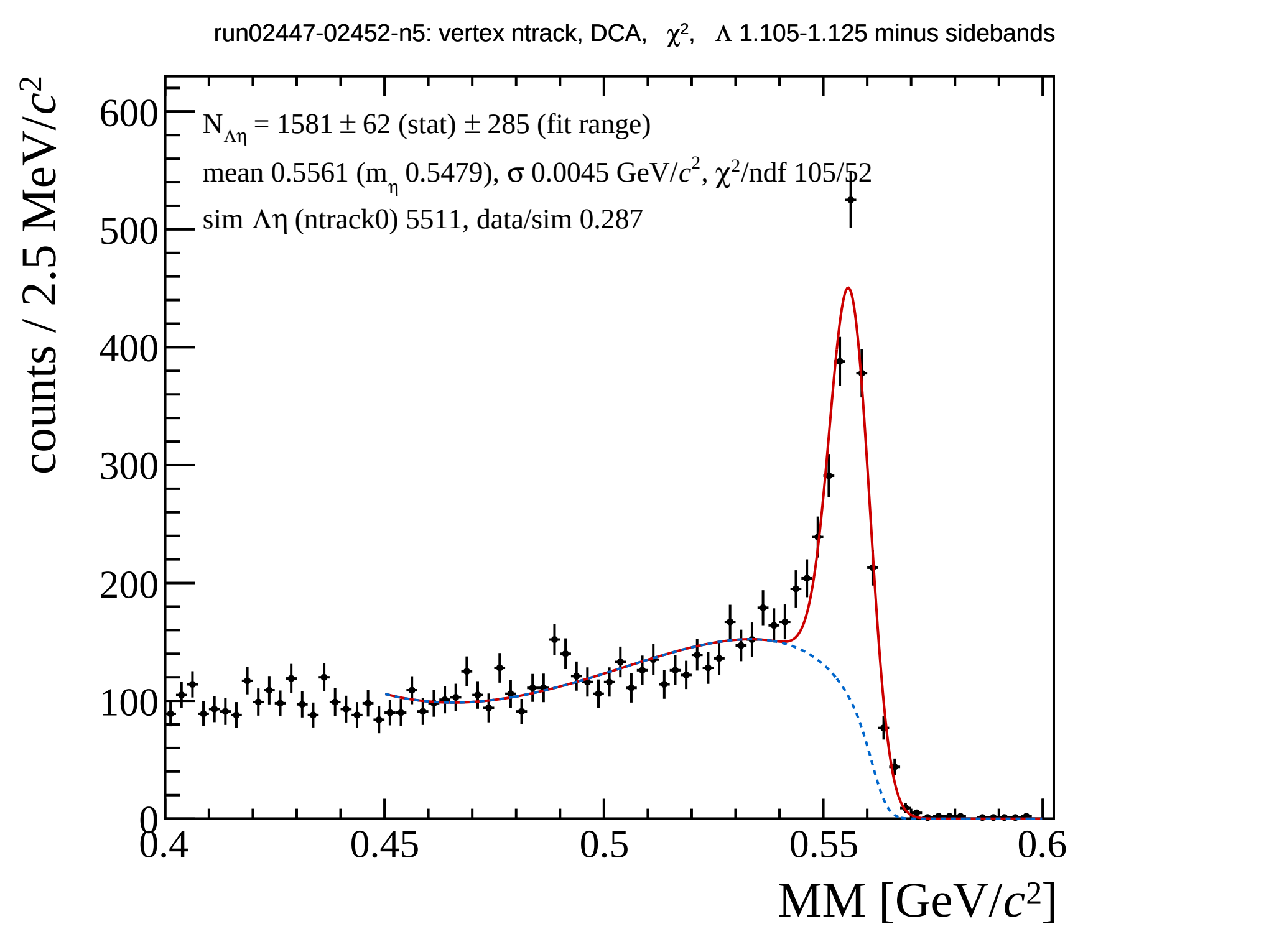

run02447-02452-n5: DCA, χ^2, LH2, kin., $m_{p\pi^-}$ 1.09-1.13
$\times 10^3$
counts / 10 MeV/c^2
1.4
1.2
1
0.8
0.6
0.4
0.2
0
−0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c^2]

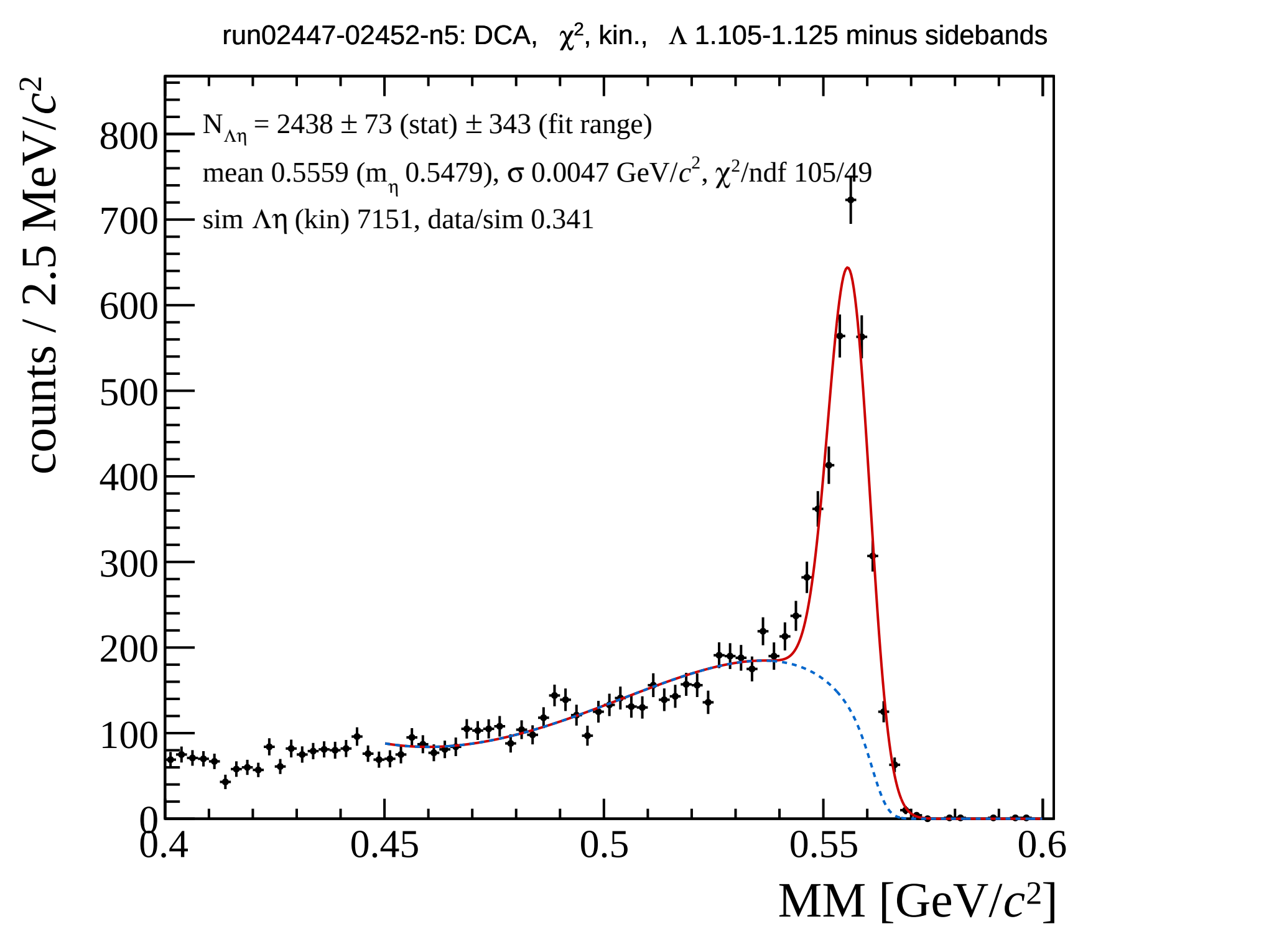

run02447-02452-n5: ntrack, DCA, χ^2, LH2, kin., $m_{p\pi^-}$ 1.09-1.13
$\times 10^3$
counts / 10 MeV/c^2
1
0.8
0.6
0.4
0.2
0
−0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c^2]

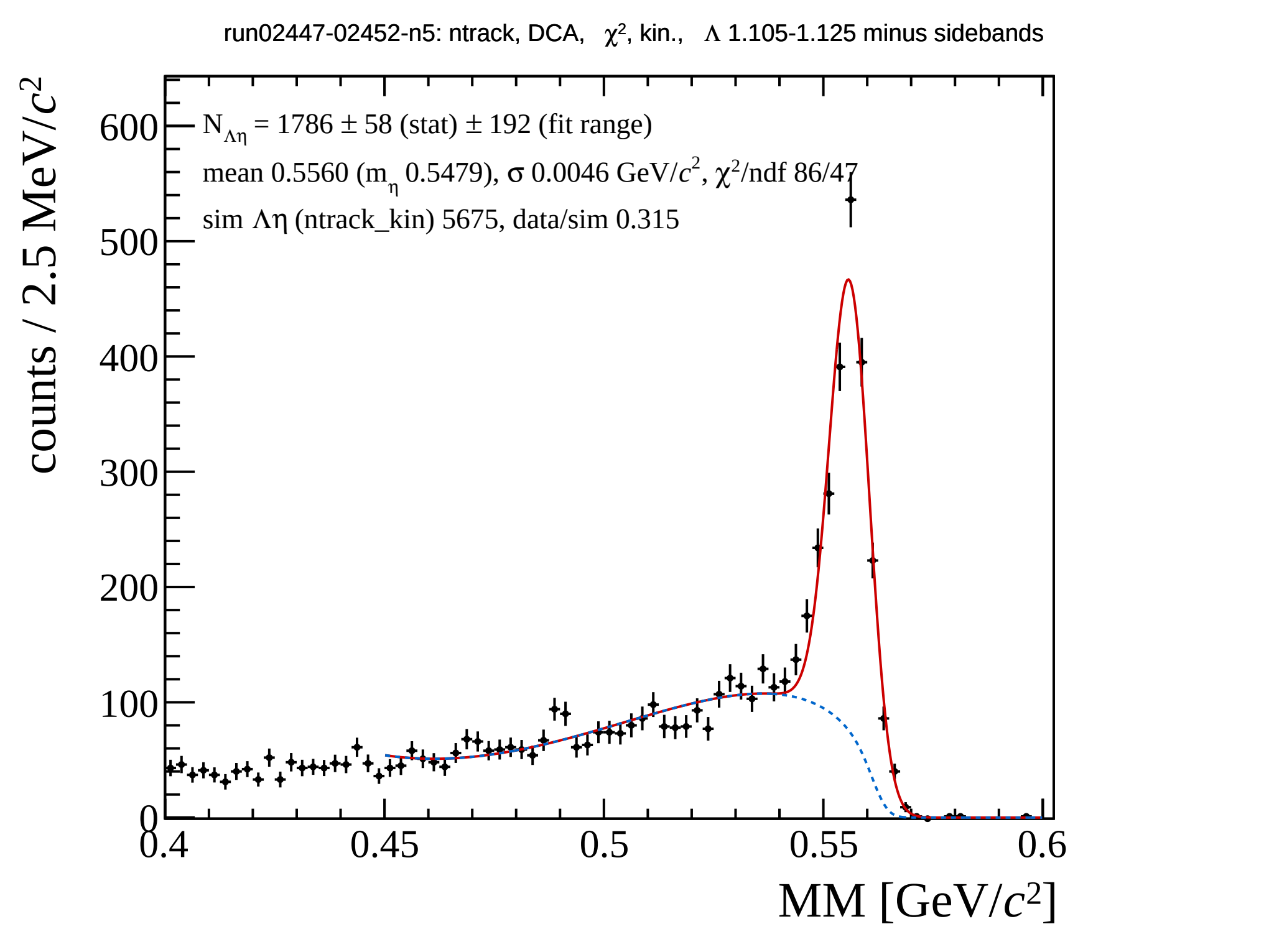

run02447-02452-n5: vertex ntrack, DCA, χ^2, LH2, kin., $m_{p\pi^-}$ 1.09-1.13
counts / 10 MeV/c^2
900
800
700
600
500
400
300
200
100
0
−0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
MM [GeV/c^2]

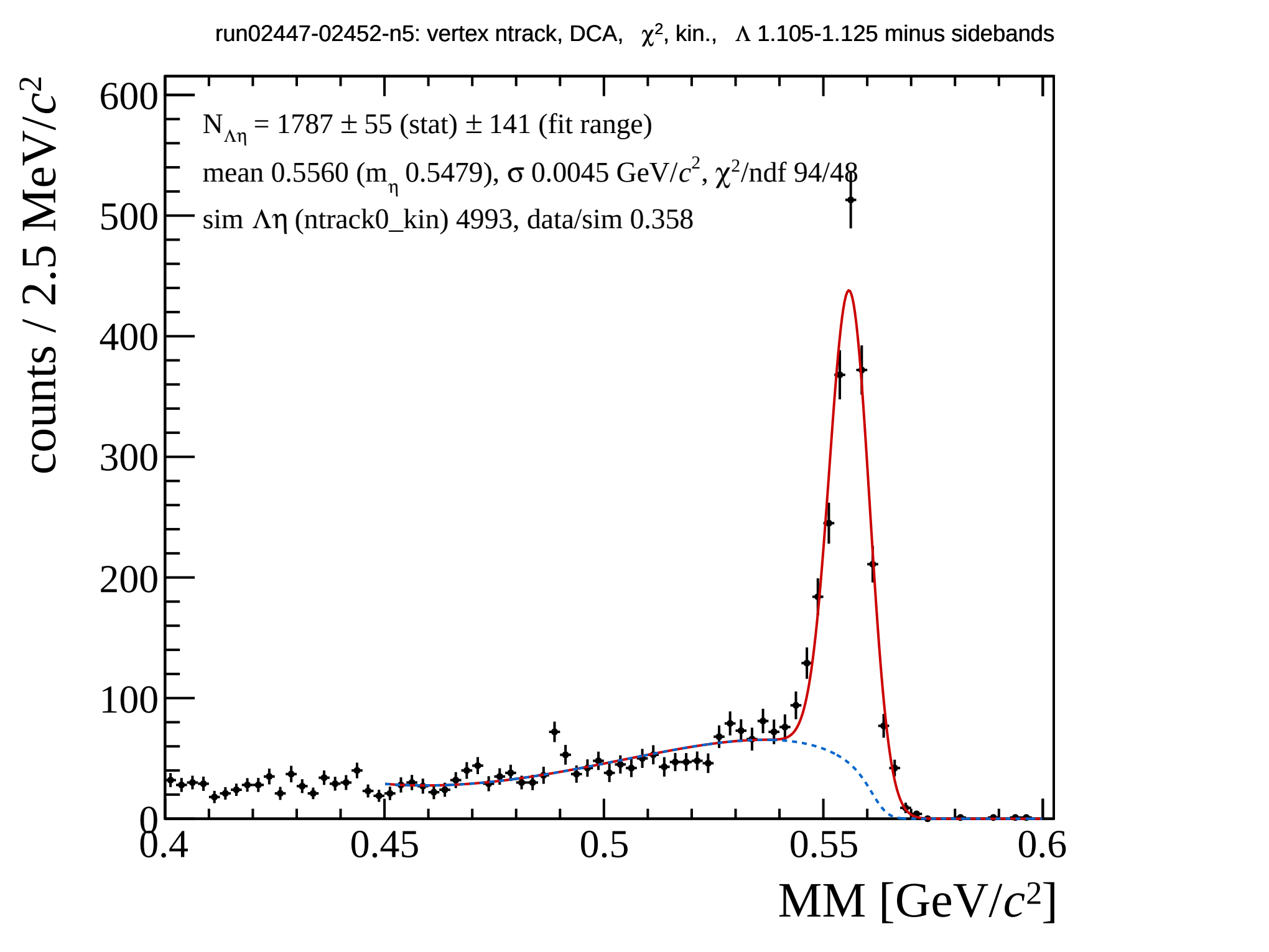

run02447-02452-n5: DCA, χ^2, Λ 1.105-1.125 minus sidebands
$N_{\Lambda\eta}$ = 2038 ± 82 (stat) ± 495 (fit range)
mean 0.5561 (m_η 0.5479), σ 0.0046 GeV/c^2, χ^2/ndf 112/52
sim Λη (tpc) 7895, data/sim 0.258
counts / 2.5 MeV/c^2
0
100
200
300
400
500
600
700
800
0.4
0.45
0.5
0.55
0.6
MM [GeV/c^2]


run02447-02452-n5: ntrack, DCA, χ², Λ 1.105-1.125 minus sidebands
N_Λη = 1521 ± 66 (stat) ± 291 (fit range)
mean 0.5562 (m_η 0.5479), σ 0.0045 GeV/c², χ²/ndf 82/49
sim Λη (ntrack) 6264, data/sim 0.243
counts / 2.5 MeV/c²
MM [GeV/c²]
0
100
200
300
400
500
600
0.4
0.45
0.5
0.55
0.6


run02447-02452-n5: vertex ntrack, DCA, χ², Λ 1.105-1.125 minus sidebands
N_Λη = 1581 ± 62 (stat) ± 285 (fit range)
mean 0.5561 (m_η 0.5479), σ 0.0045 GeV/c², χ²/ndf 105/52
sim Λη (ntrack0) 5511, data/sim 0.287
counts / 2.5 MeV/c²
MM [GeV/c²]
0
100
200
300
400
500
600
0.4
0.45
0.5
0.55
0.6


run02447-02452-n5: DCA, χ^2, kin., Λ 1.105-1.125 minus sidebands
$N_{\Lambda\eta}$ = 2438 ± 73 (stat) ± 343 (fit range)
mean 0.5559 (m_η 0.5479), σ 0.0047 GeV/c^2, χ^2/ndf 105/49
sim Λη (kin) 7151, data/sim 0.341
counts / 2.5 MeV/c^2
0
100
200
300
400
500
600
700
800
0.4
0.45
0.5
0.55
0.6
MM [GeV/c^2]


run02447-02452-n5: ntrack, DCA, χ², kin., Λ 1.105-1.125 minus sidebands
N_Λη = 1786 ± 58 (stat) ± 192 (fit range)
mean 0.5560 (m_η 0.5479), σ 0.0046 GeV/c², χ²/ndf 86/47
sim Λη (ntrack_kin) 5675, data/sim 0.315
counts / 2.5 MeV/c²
MM [GeV/c²]
0
100
200
300
400
500
600
0.4
0.45
0.5
0.55
0.6


run02447-02452-n5: vertex ntrack, DCA, χ², kin., Λ 1.105-1.125 minus sidebands
N_Λη = 1787 ± 55 (stat) ± 141 (fit range)
mean 0.5560 (m_η 0.5479), σ 0.0045 GeV/c², χ²/ndf 94/48
sim Λη (ntrack0_kin) 4993, data/sim 0.358
counts / 2.5 MeV/c²
MM [GeV/c²]
0
100
200
300
400
500
600
0.4
0.45
0.5
0.55
0.6

Lambda eta yield, run02447-02452-n5

fit of X_mass (Lambda 1.105-1.125 minus sidebands) in 0.45-0.60 GeV/c2:

Gaussian eta + (quadratic) x smeared sqrt edge at the kinematic endpoint

endpoint 0.5625, smearing 0.0025 GeV/c2 from the first selection, fixed in the others

systematic: largest change with the lower fit edge at 0.40 or 0.48

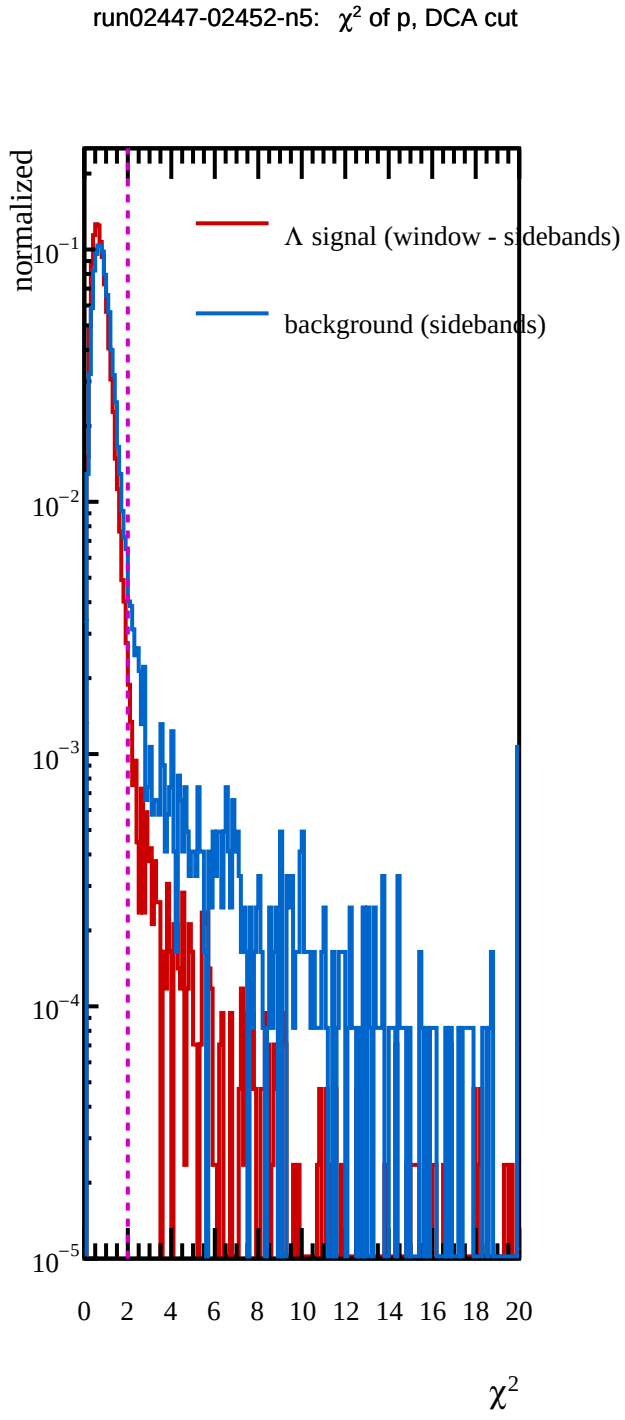
sim: Lambda_eta row of yield-sim.root (true kinematics, x eff_tpc_L), same condition

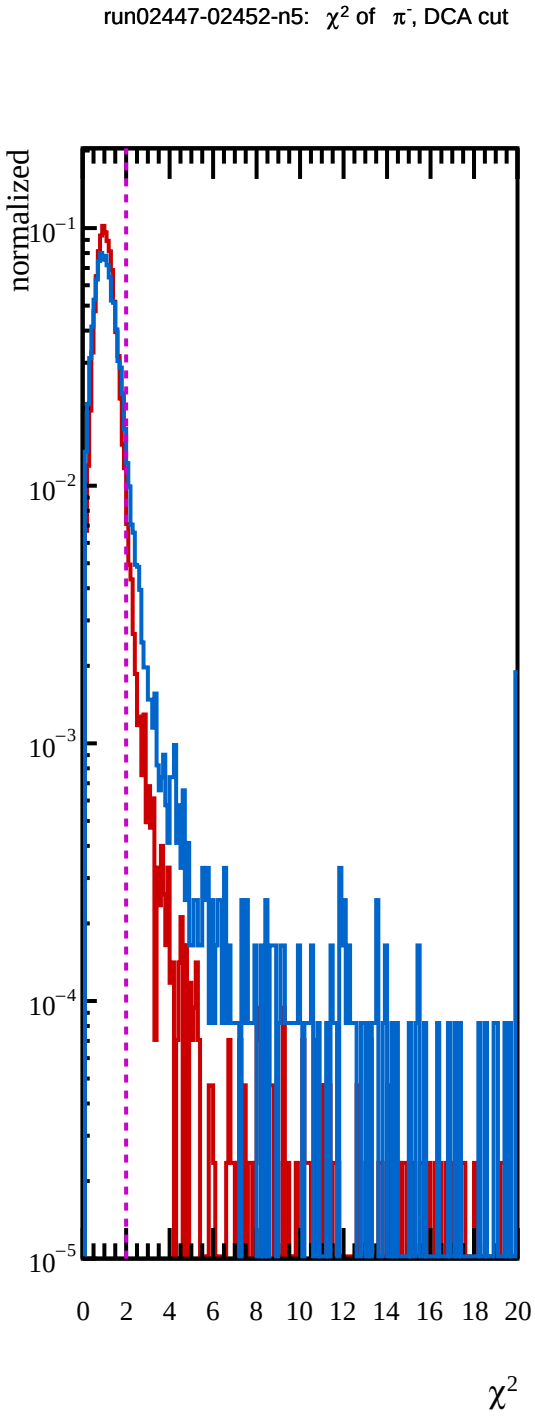
all 735 MeV/c runs: these runs x (K-Beam x DAQ-Eff of all / these) = x 19.38;

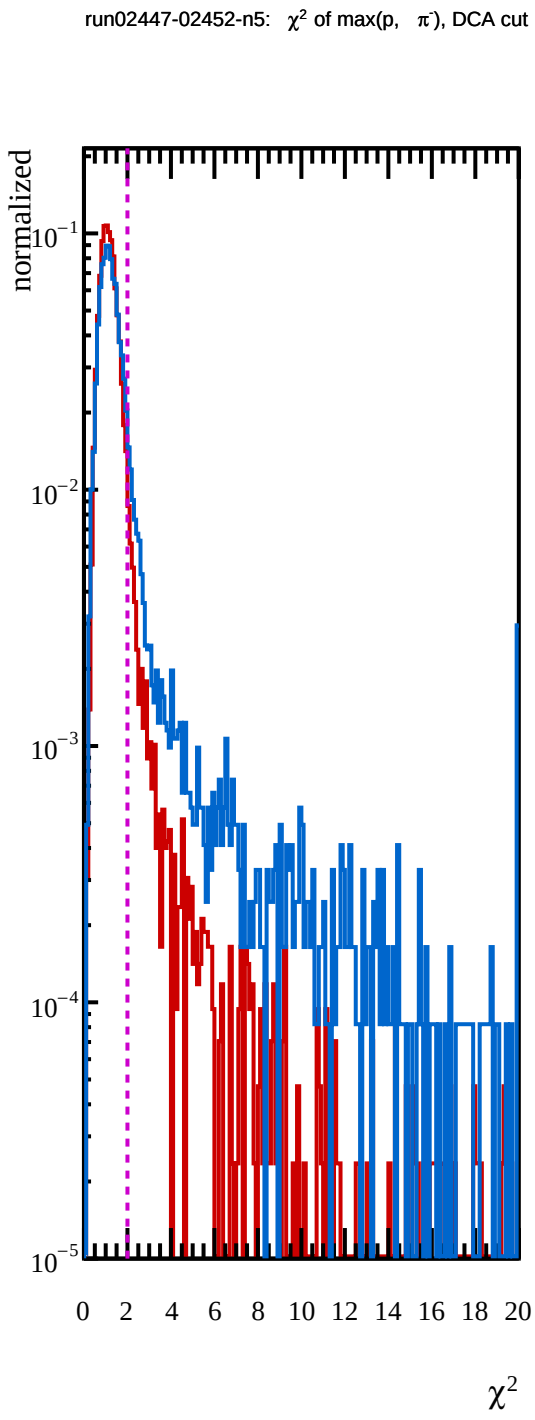
stat x sqrt of that ratio, syst and sim x the ratio (runs from kbeam_statistics.txt)

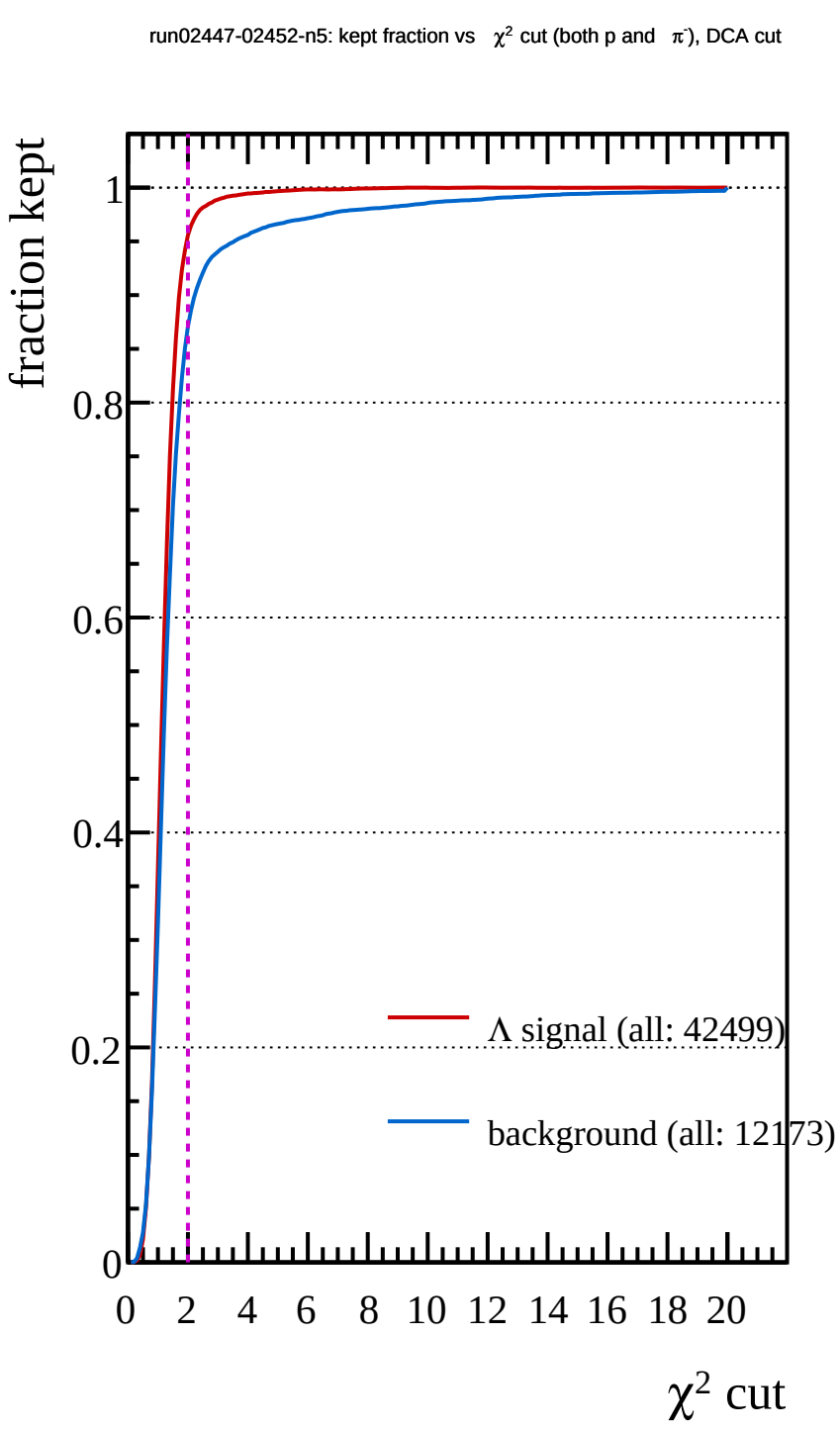
	runs 2447-2452	all 735 MeV/c	sim	sim all	data/sim
runs	5	141			
K-Beam	6.835e+07	1.324e+09			
K-Beam x DAQ-Eff	5.14e+07	9.963e+08			
N_Lambda_eta +- stat +- syst					
DCA, chi2	2038 +- 82 +- 495	39501 +- 359 +- 9593	7895	153010	0.258
ntrack, DCA, chi2	1521 +- 66 +- 291	29488 +- 289 +- 5639	6264	121400	0.243
vertex ntrack, DCA, chi2	1581 +- 62 +- 285	30643 +- 274 +- 5524	5511	106819	0.287
DCA, chi2, kin.	2438 +- 73 +- 343	47250 +- 321 +- 6645	7151	138589	0.341
ntrack, DCA, chi2, kin.	1786 +- 58 +- 192	34614 +- 257 +- 3722	5675	109981	0.315
vertex ntrack, DCA, chi2, kin.	1787 +- 55 +- 141	34642 +- 240 +- 2731	4993	96767	0.358

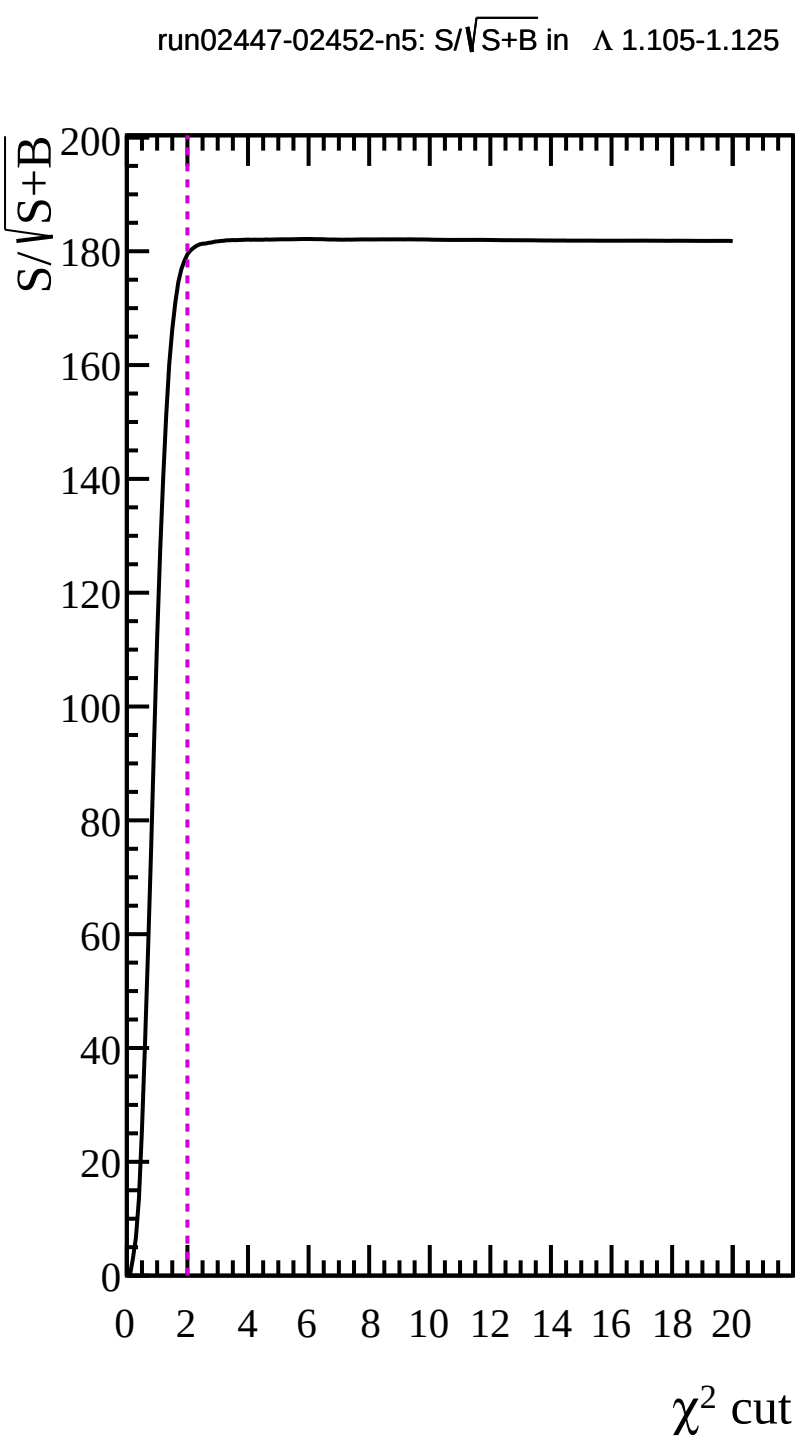
sim column (yield-sim.root): tpc, ntrack, ntrack0, kin, ntrack_kin, ntrack0_kin in this order


run02447-02452-n5: χ^2 of p, DCA cut
normalized
Λ signal (window - sidebands)
background (sidebands)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0 2 4 6 8 10 12 14 16 18 20
χ^2


run02447-02452-n5: χ^2 of π^-, DCA cut
normalized
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0 2 4 6 8 10 12 14 16 18 20
χ^2


run02447-02452-n5: χ^2 of max(p, π^-), DCA cut
normalized
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0 2 4 6 8 10 12 14 16 18 20
χ^2


run02447-02452-n5: kept fraction vs χ^2 cut (both p and π^-), DCA cut
fraction kept
1
0.8
0.6
0.4
0.2
0
0 2 4 6 8 10 12 14 16 18 20
χ^2 cut
Λ signal (all: 42499)
background (all: 12173)


run02447-02452-n5: $S/\sqrt{S+B}$ in Λ 1.105-1.125
$S/\sqrt{S+B}$
200
180
160
140
120
100
80
60
40
20
0
0 2 4 6 8 10 12 14 16 18 20
χ^2 cut

Lambda yield vs cut, run02447-02452-n5 (N_sig, N_bkg in 1.09-1.13 GeV/c2)

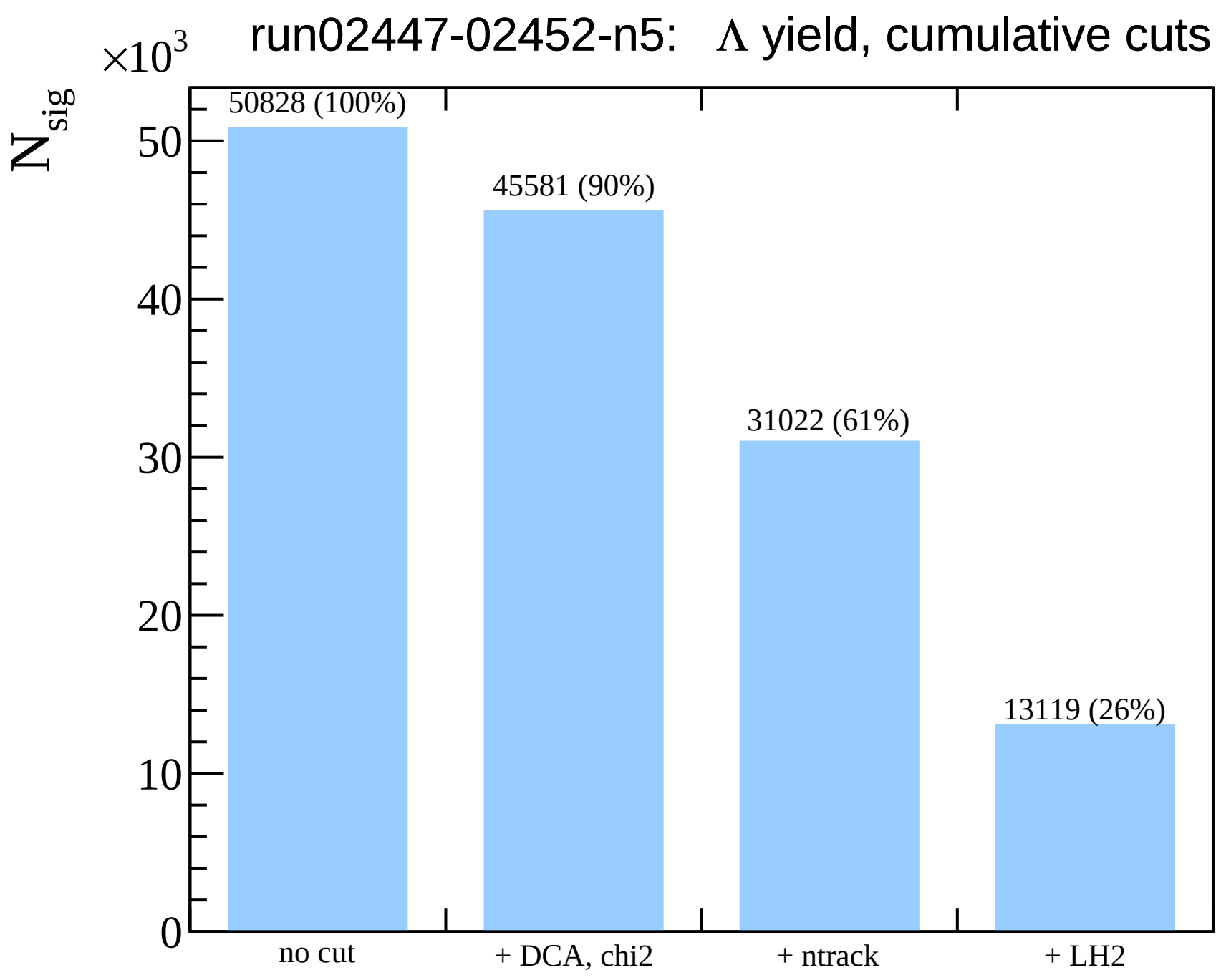
Each cut alone, one Lambda per event	N_sig	/nocut	N_bkg	/nocut
no cut	50828	1.000	42626	1.000
ntrack	35423	0.697	28718	0.674
vertex ntrack	27194	0.535	23051	0.541
DCA < 5 mm	47365	0.932	29287	0.687
chi2 <= 2 (p and pi)	48973	0.964	36518	0.857
LH2 production vertex	21169	0.416	12053	0.283

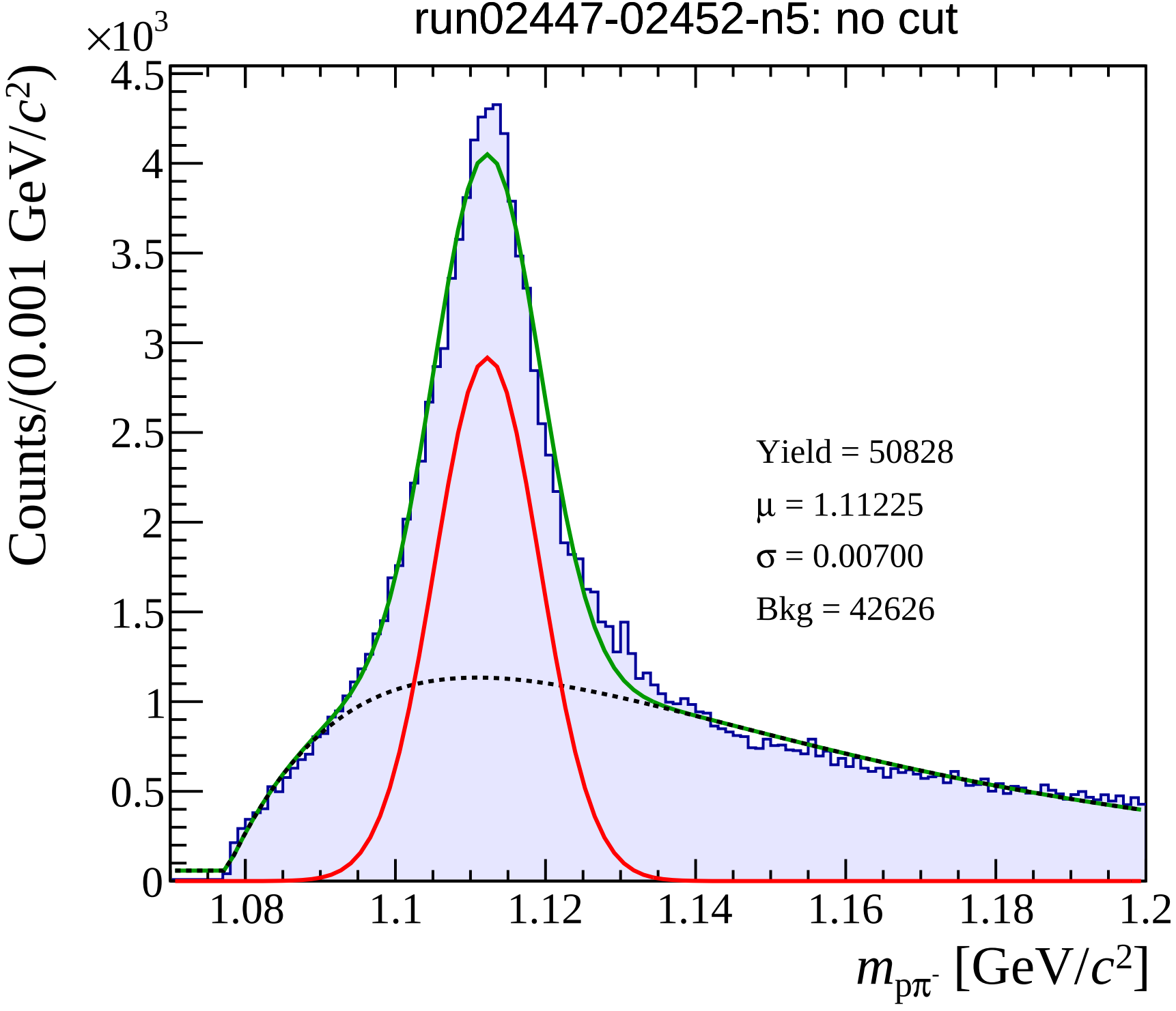
Cumulative (step ratio)	N_sig	/prev	N_bkg	/prev
no cut	50828	1.000	42626	1.000
+ DCA, chi2	45581	0.897	25356	0.595
+ ntrack	31022	0.681	17279	0.681
+ LH2	13119	0.423	6138	0.355
total (last / first)		0.258		0.144

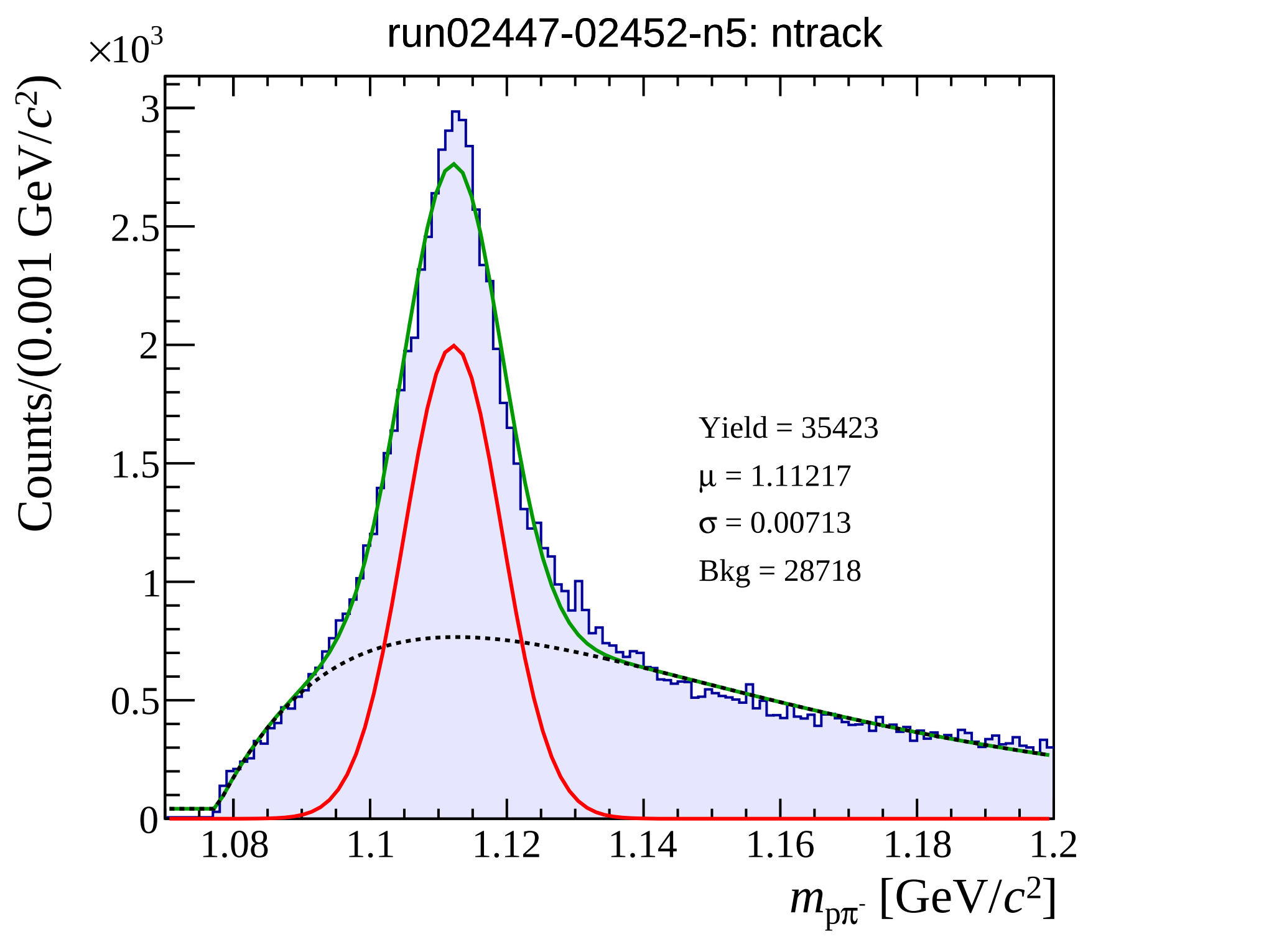
Vertex ntrack instead of ntrack (step ratio)				
+ DCA, chi2	45581	0.897	25356	0.595
+ vertex ntrack	23942	0.525	12825	0.506
+ LH2	10516	0.439	5000	0.390
vertex ntrack / ntrack	23942	0.772	12825	0.742
vertex ntrack / ntrack, LH2	10516	0.802	5000	0.814
vertex ntrack / ntrack, LH2, MM	2377	0.743	356	0.654

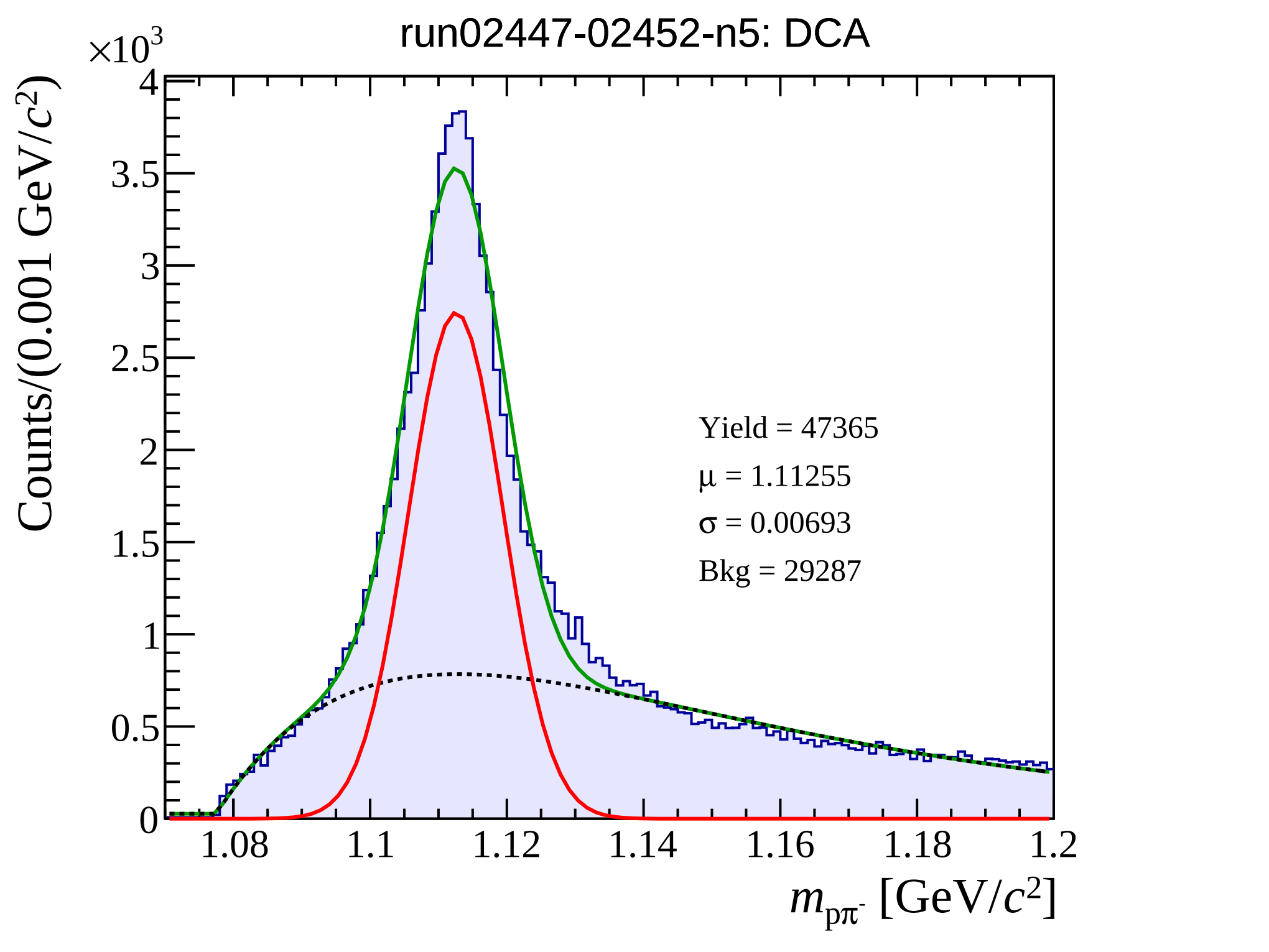
Kinematics: p_pi < 0.25, 0.4 <= p_Lambda <= 0.6 GeV/c (step ratio)				
DCA, chi2, LH2 + kin.	6672	0.346	1558	0.194
ntrack, DCA, chi2, LH2 + kin.	3931	0.300	1321	0.215
vtx ntrack, DCA, chi2, LH2 + kin.	3337	0.317	591	0.118
DCA, chi2, LH2, MM + kin.	3273	0.634	646	0.839
ntrack, DCA, chi2, LH2, MM + kin.	1960	0.613	516	0.948
vtx ntrack, ..., LH2, MM + kin.	1477	0.621	317	0.890

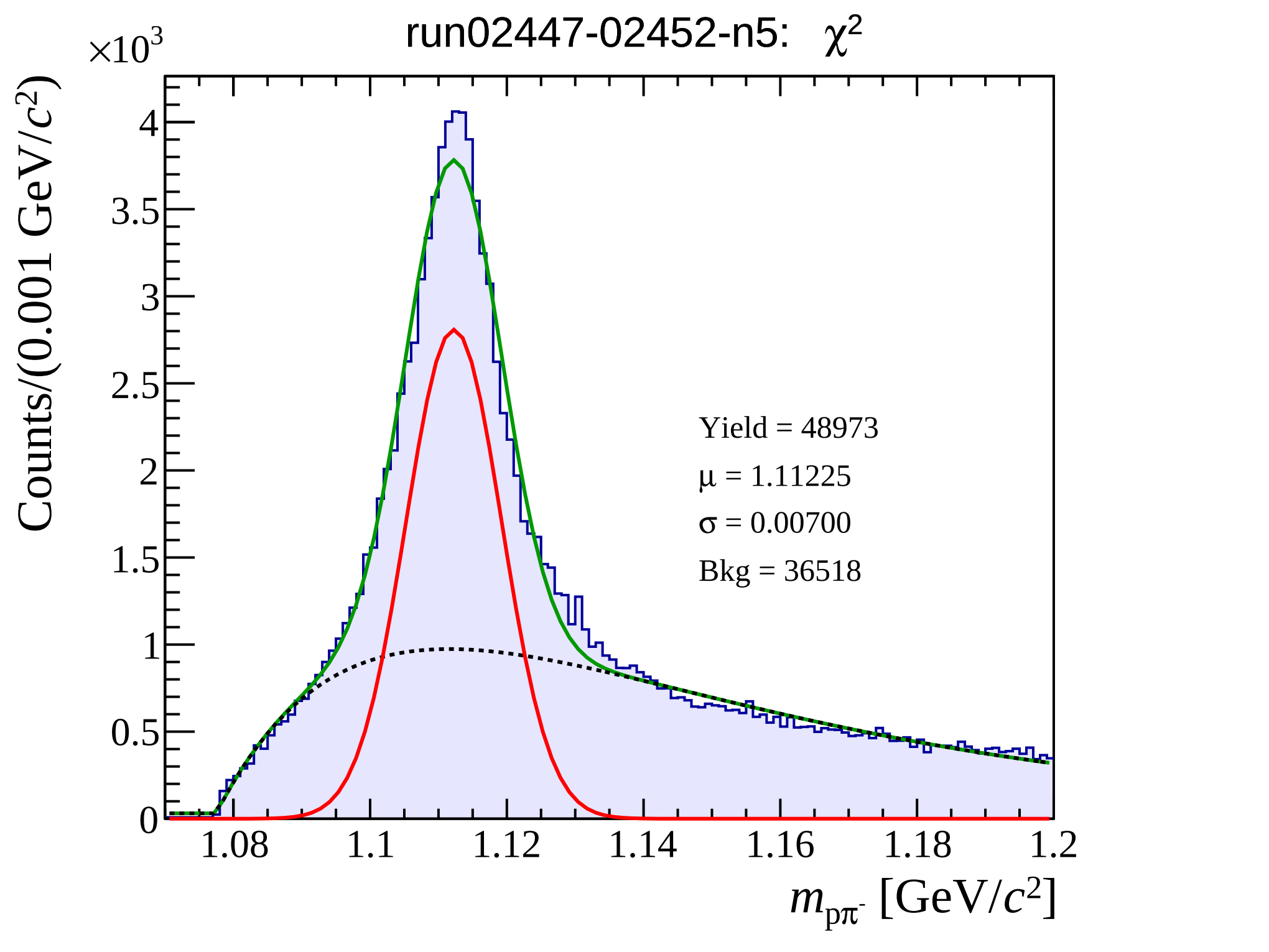
(the cut_flag kinematics digit also holds the mass window 1.09-1.12; not used)

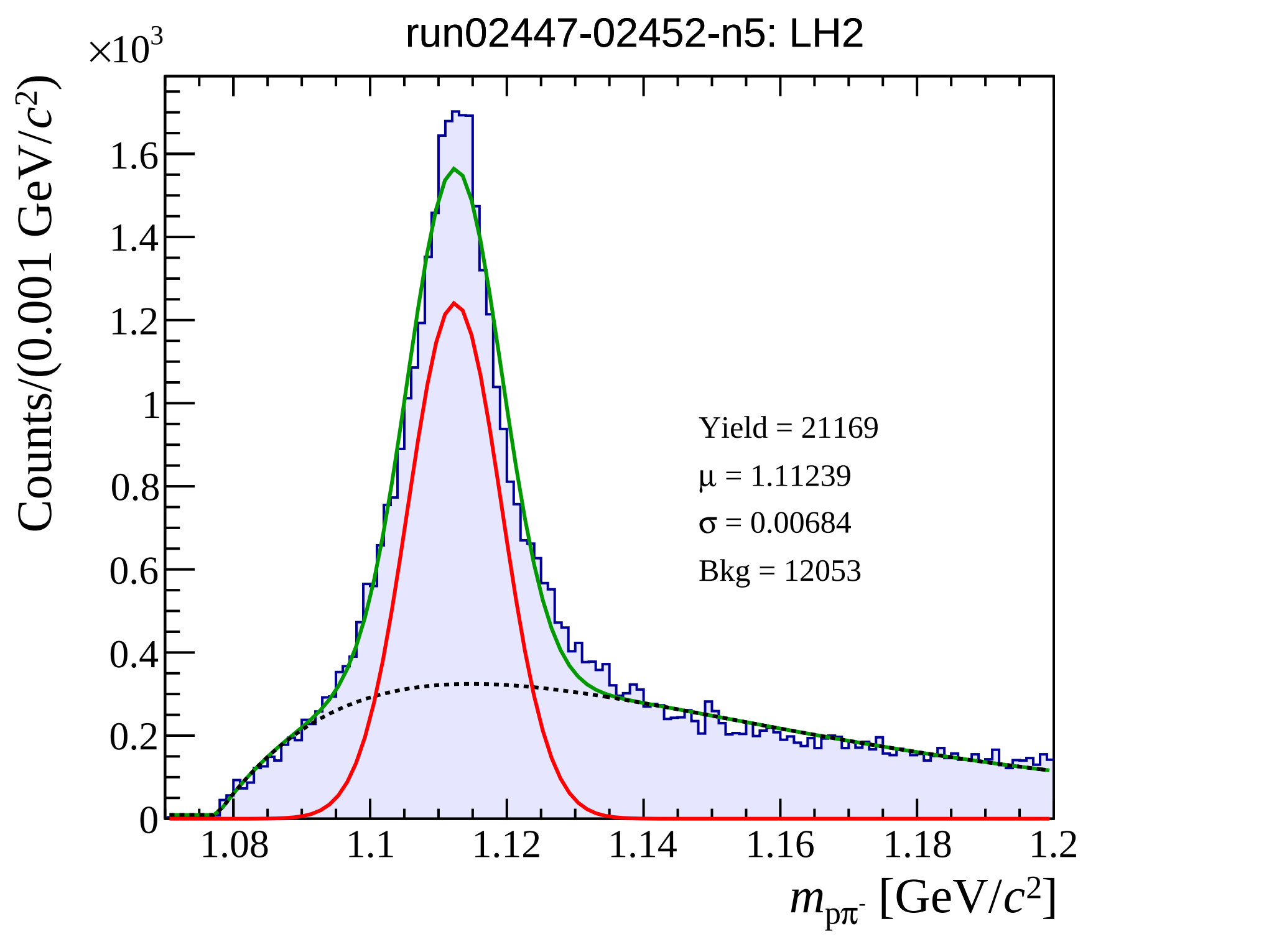

run02447-02452-n5: Λ yield, cumulative cuts
×10^3
N_sig
50828 (100%)
45581 (90%)
31022 (61%)
13119 (26%)
0
10
20
30
40
50
no cut
+ DCA, chi2
+ ntrack
+ LH2

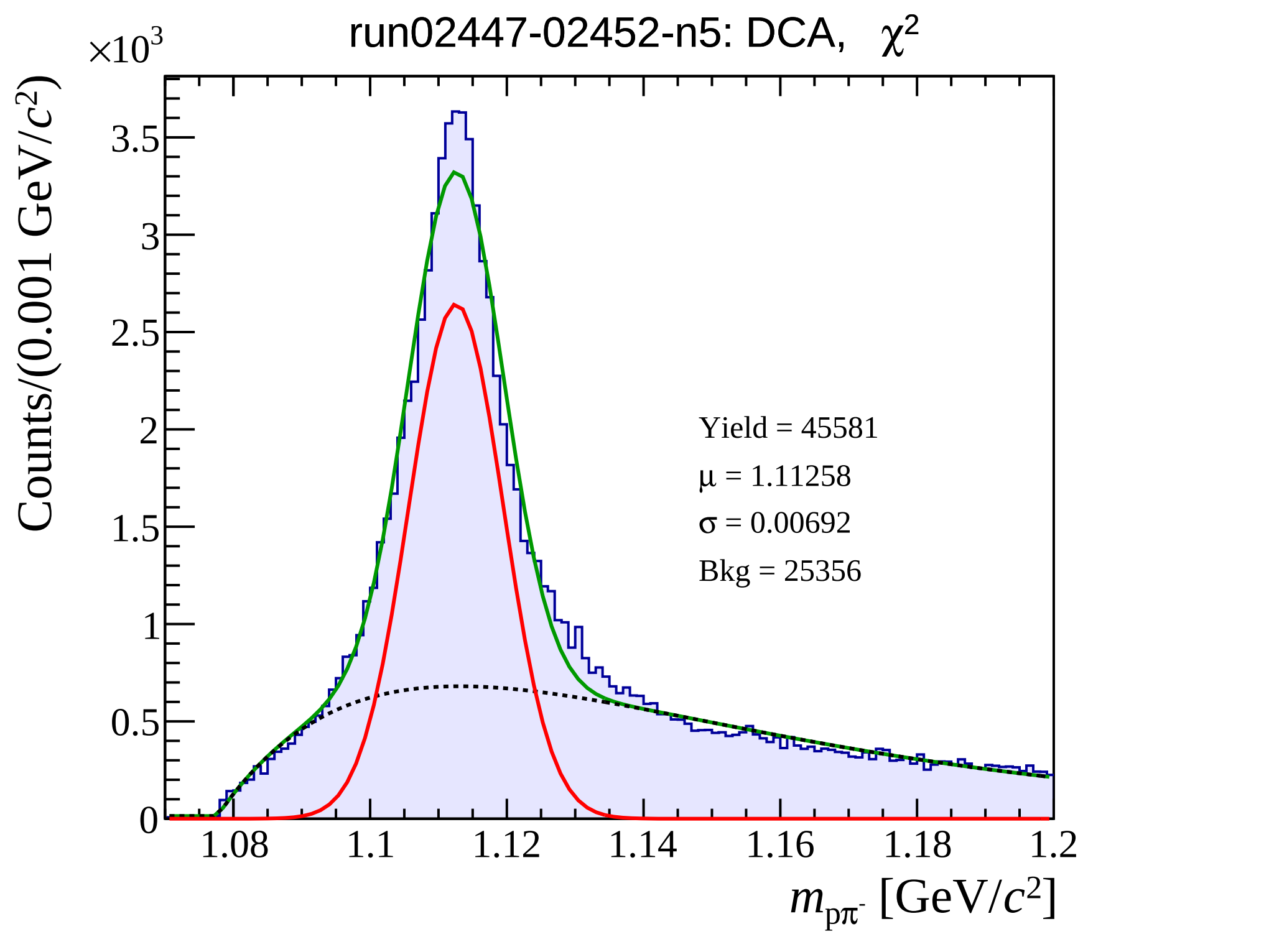

run02447-02452-n5: no cut
×10^3
Counts/(0.001 GeV/c²)
4.5
4
3.5
3
2.5
2
1.5
1
0.5
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m_pπ- [GeV/c²]
Yield = 50828
μ = 1.11225
σ = 0.00700
Bkg = 42626

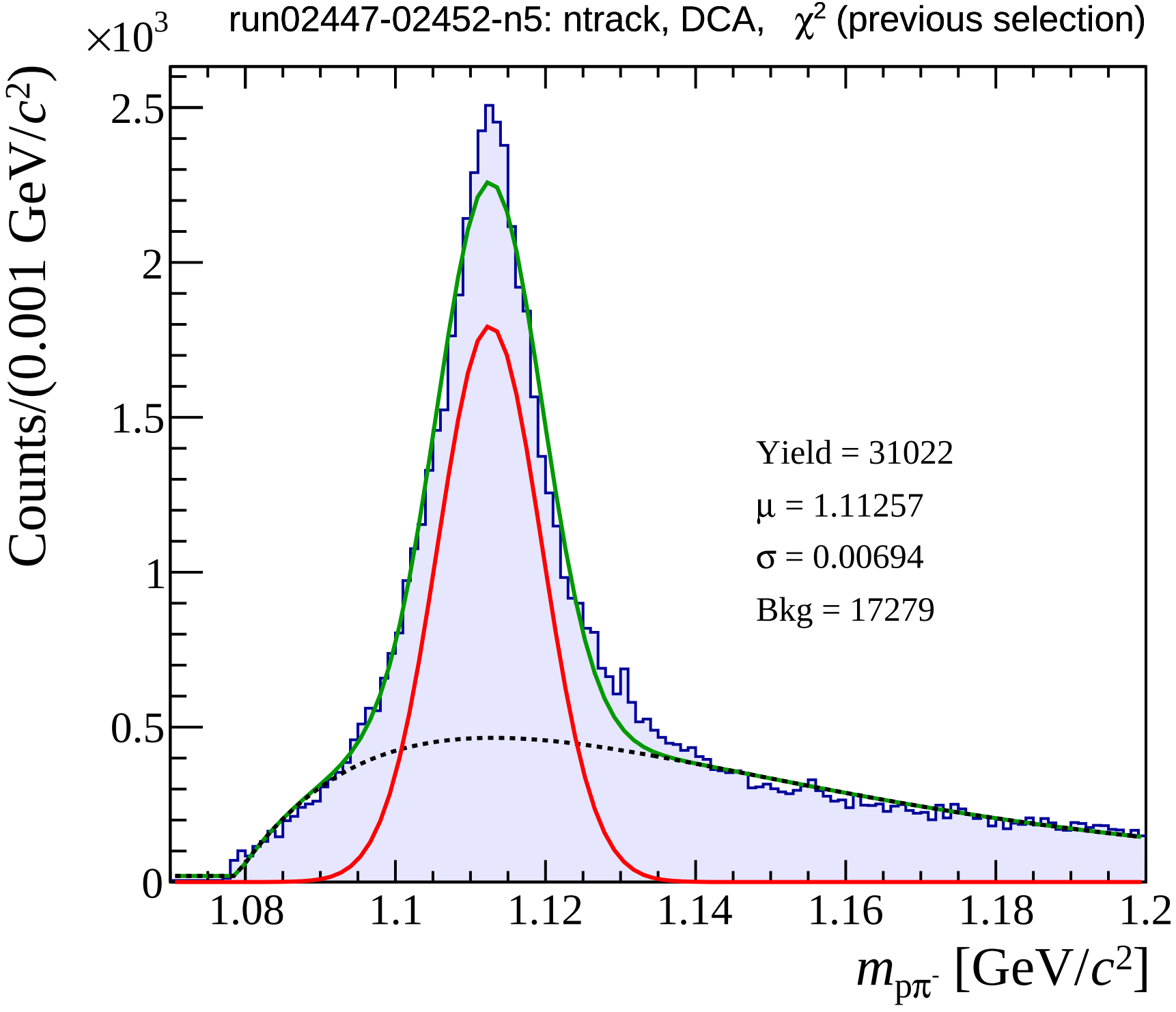

run02447-02452-n5: ntrack
×10^3
Counts/(0.001 GeV/c²)
3
2.5
2
1.5
1
0.5
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m_pπ- [GeV/c²]
Yield = 35423
μ = 1.11217
σ = 0.00713
Bkg = 28718

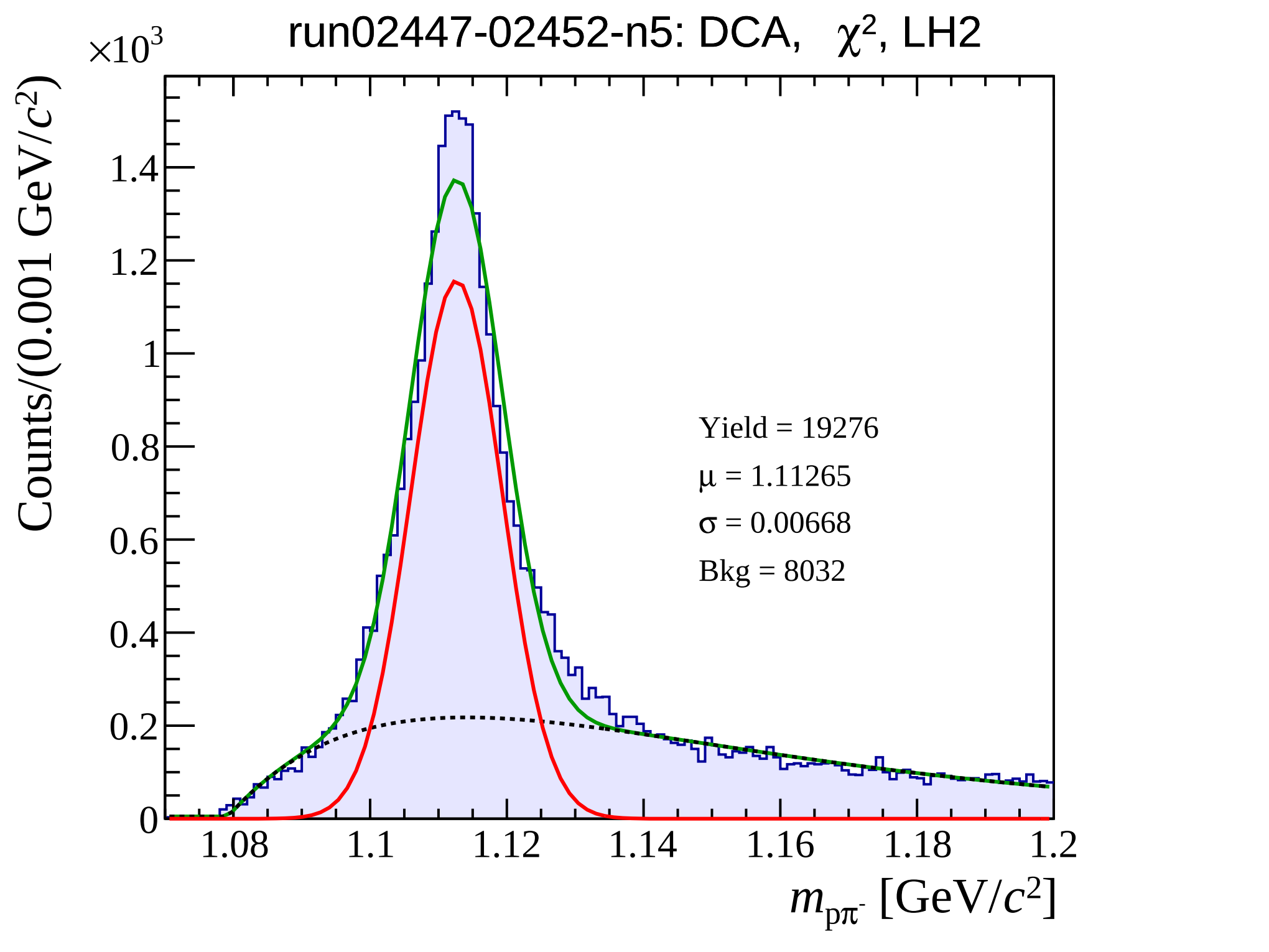

run02447-02452-n5: DCA
×10^3
Counts/(0.001 GeV/c²)
4
3.5
3
2.5
2
1.5
1
0.5
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m_pπ- [GeV/c²]
Yield = 47365
μ = 1.11255
σ = 0.00693
Bkg = 29287

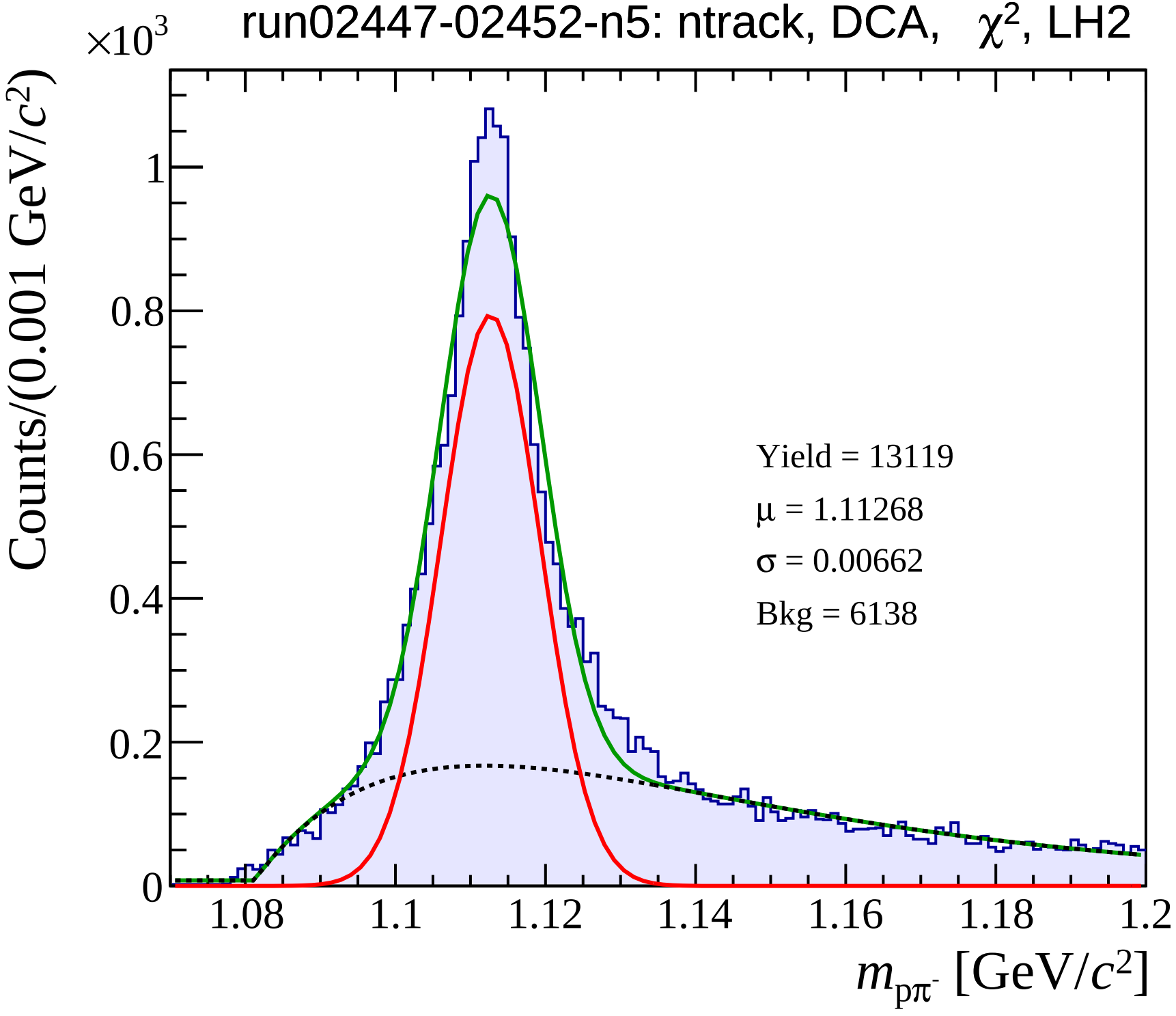

run02447-02452-n5: χ^2
$\times 10^3$
Counts/(0.001 GeV/c^2)
4
3.5
3
2.5
2
1.5
1
0.5
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 48973
μ = 1.11225
σ = 0.00700
Bkg = 36518

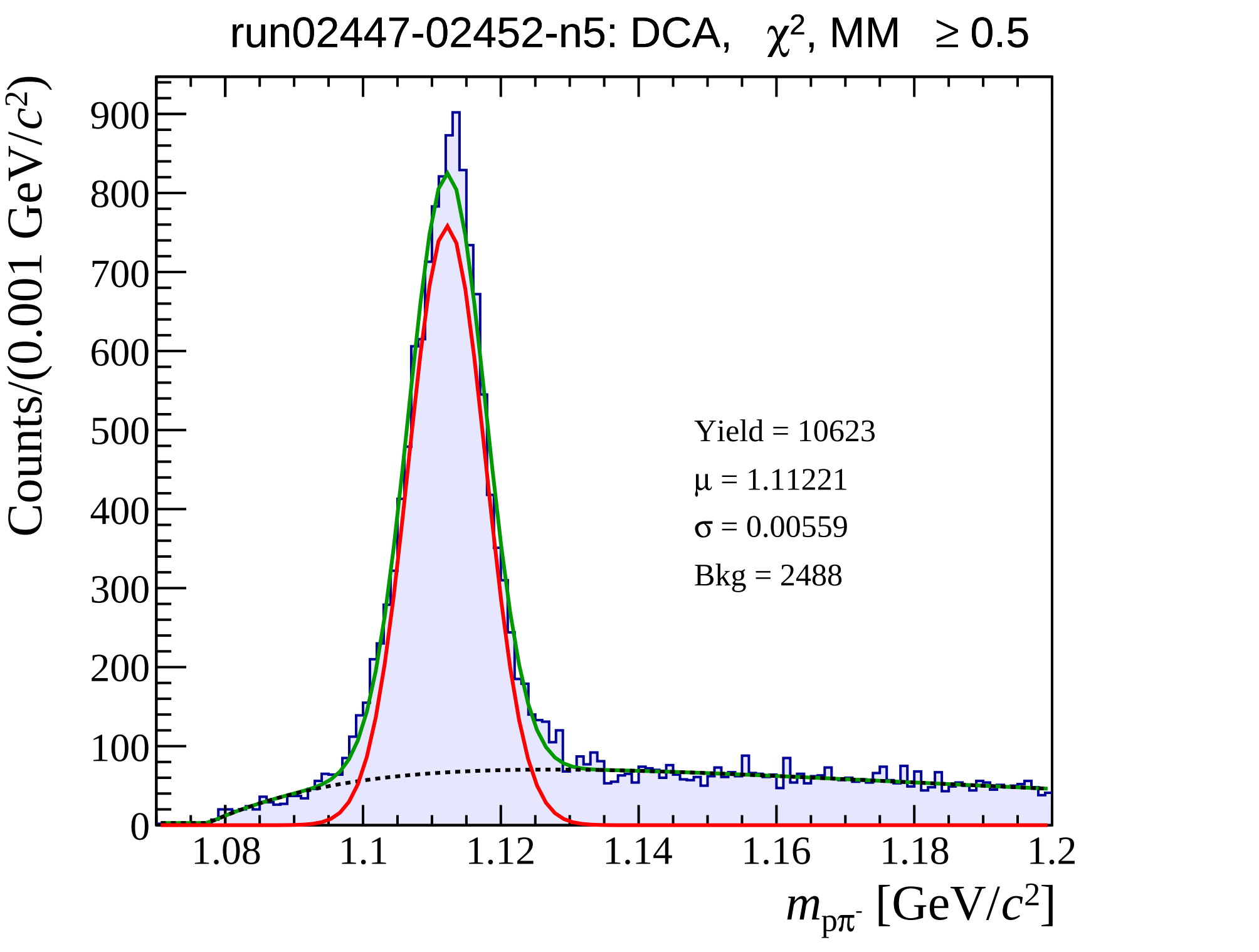

run02447-02452-n5: LH2
×10^3
Counts/(0.001 GeV/c^2)
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m_pπ- [GeV/c^2]
Yield = 21169
μ = 1.11239
σ = 0.00684
Bkg = 12053

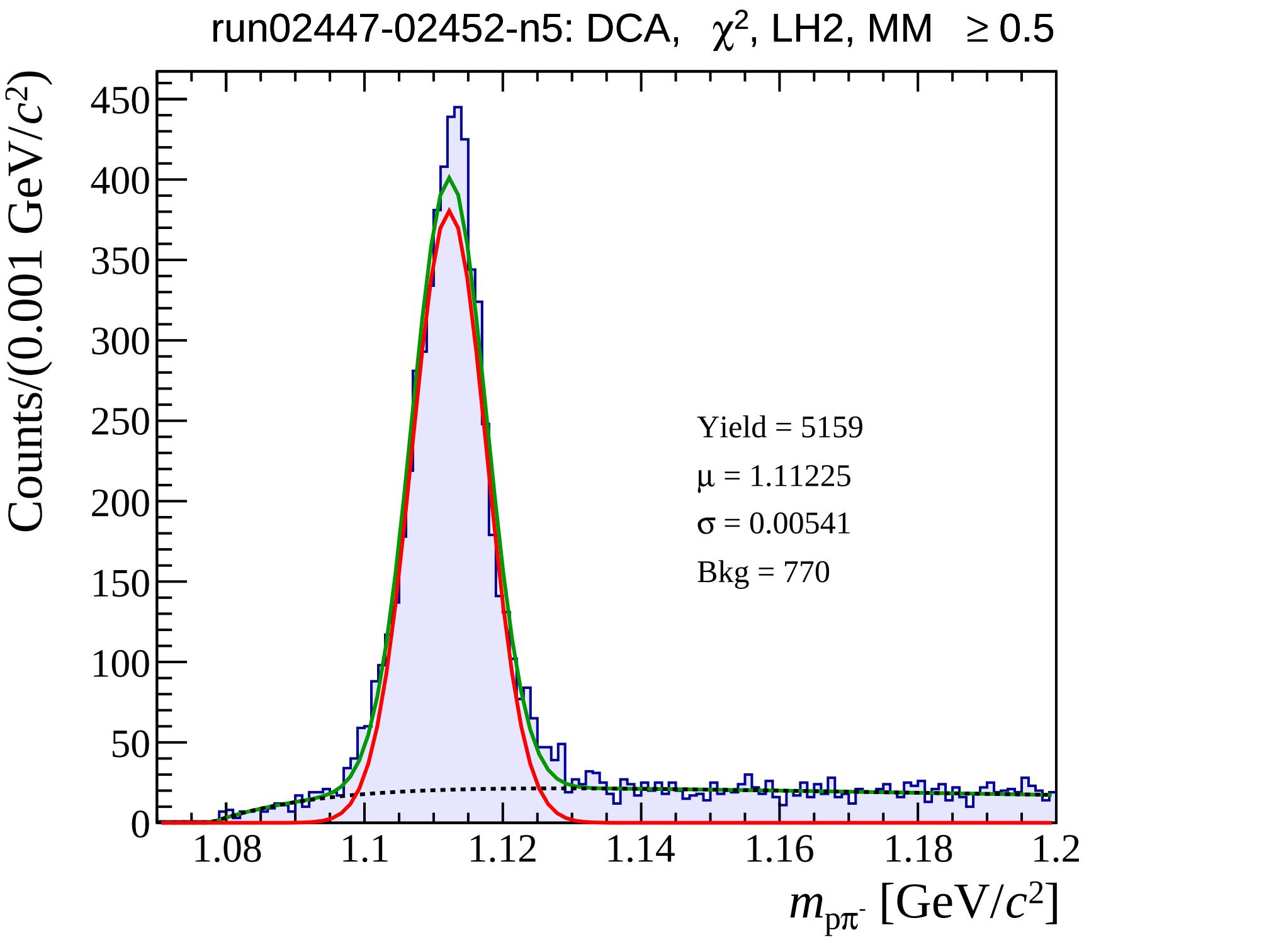

run02447-02452-n5: DCA, χ^2
$\times 10^3$
Counts/(0.001 GeV/c^2)
3.5
3
2.5
2
1.5
1
0.5
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 45581
μ = 1.11258
σ = 0.00692
Bkg = 25356

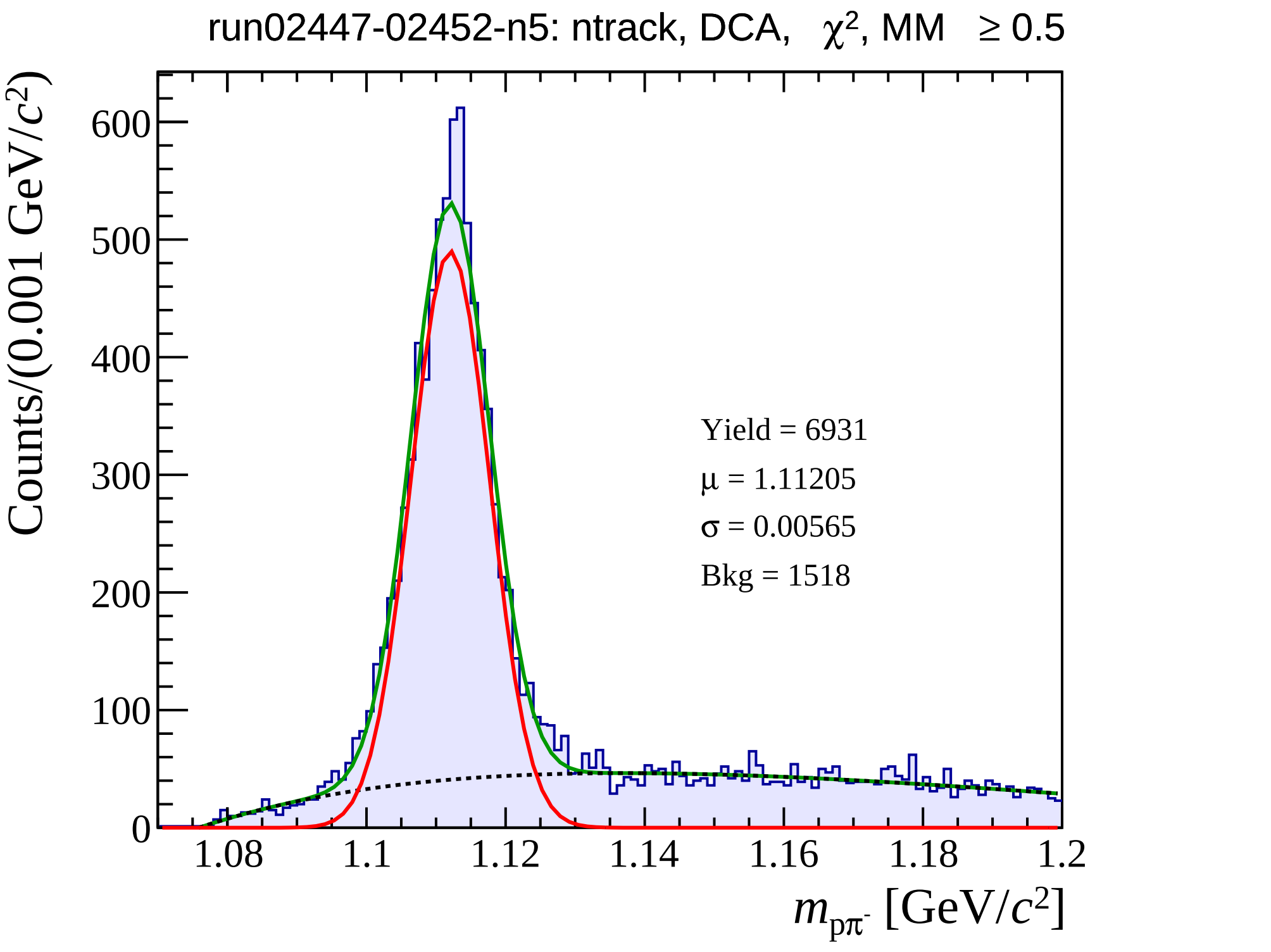

run02447-02452-n5: ntrack, DCA, χ^2 (previous selection)
$\times 10^3$
Counts/(0.001 GeV/c^2)
2.5
2
1.5
1
0.5
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 31022
μ = 1.11257
σ = 0.00694
Bkg = 17279

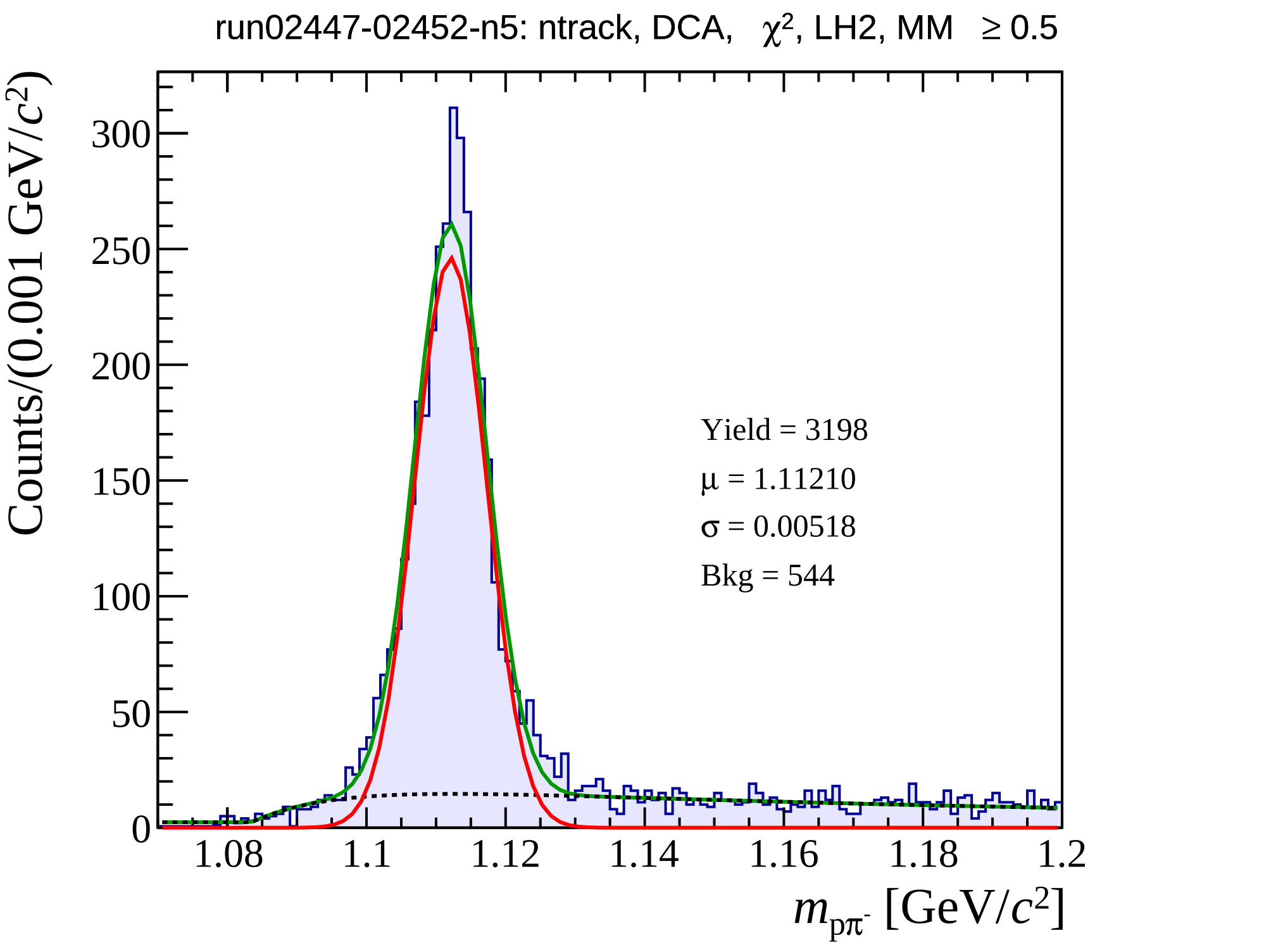

run02447-02452-n5: DCA, χ², LH2
×10³
Counts/(0.001 GeV/c²)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m pπ- [GeV/c²]
Yield = 19276
μ = 1.11265
σ = 0.00668
Bkg = 8032

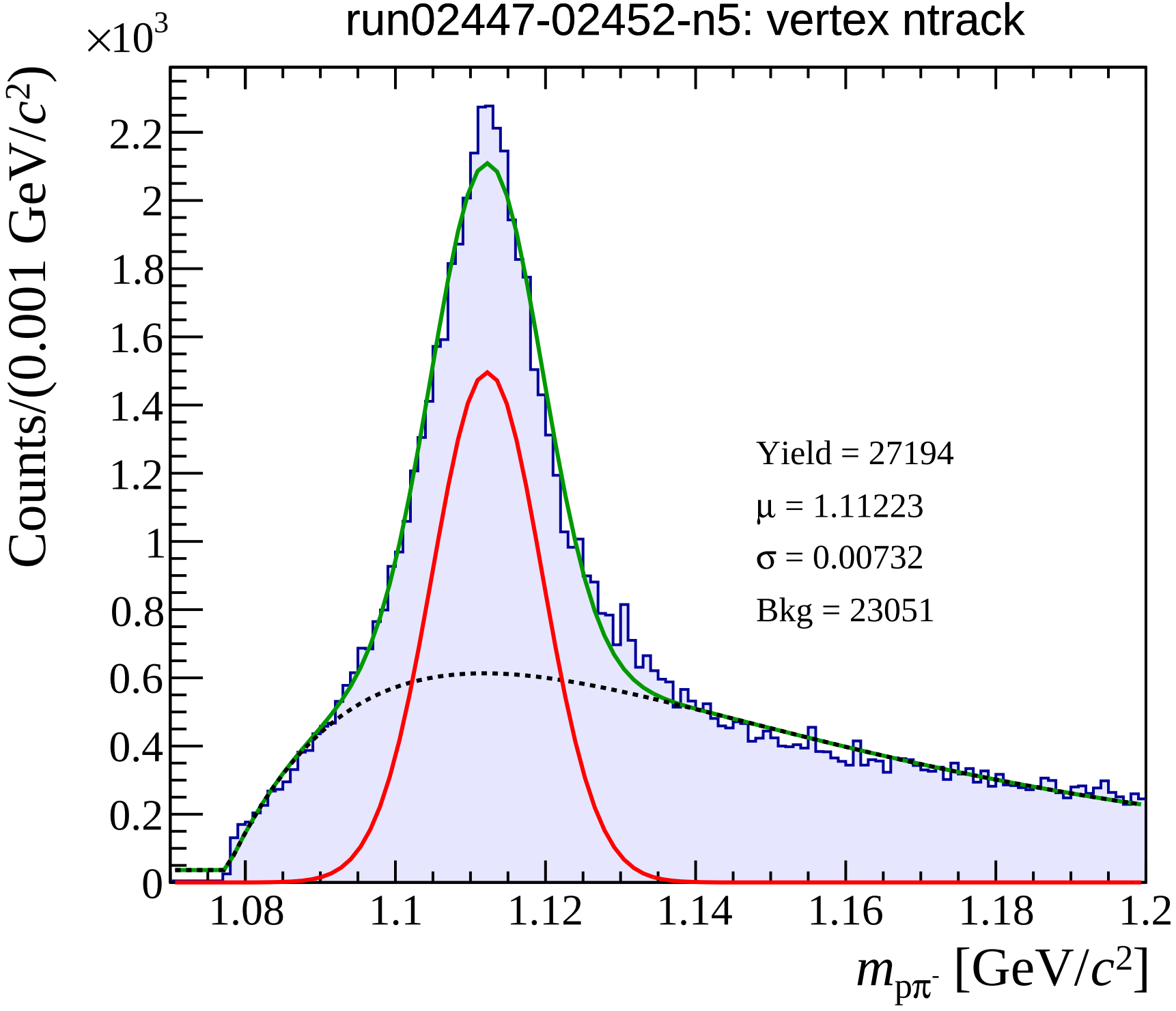

run02447-02452-n5: ntrack, DCA, χ², LH2
×10³
Counts/(0.001 GeV/c²)
1
0.8
0.6
0.4
0.2
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m pπ- [GeV/c²]
Yield = 13119
μ = 1.11268
σ = 0.00662
Bkg = 6138

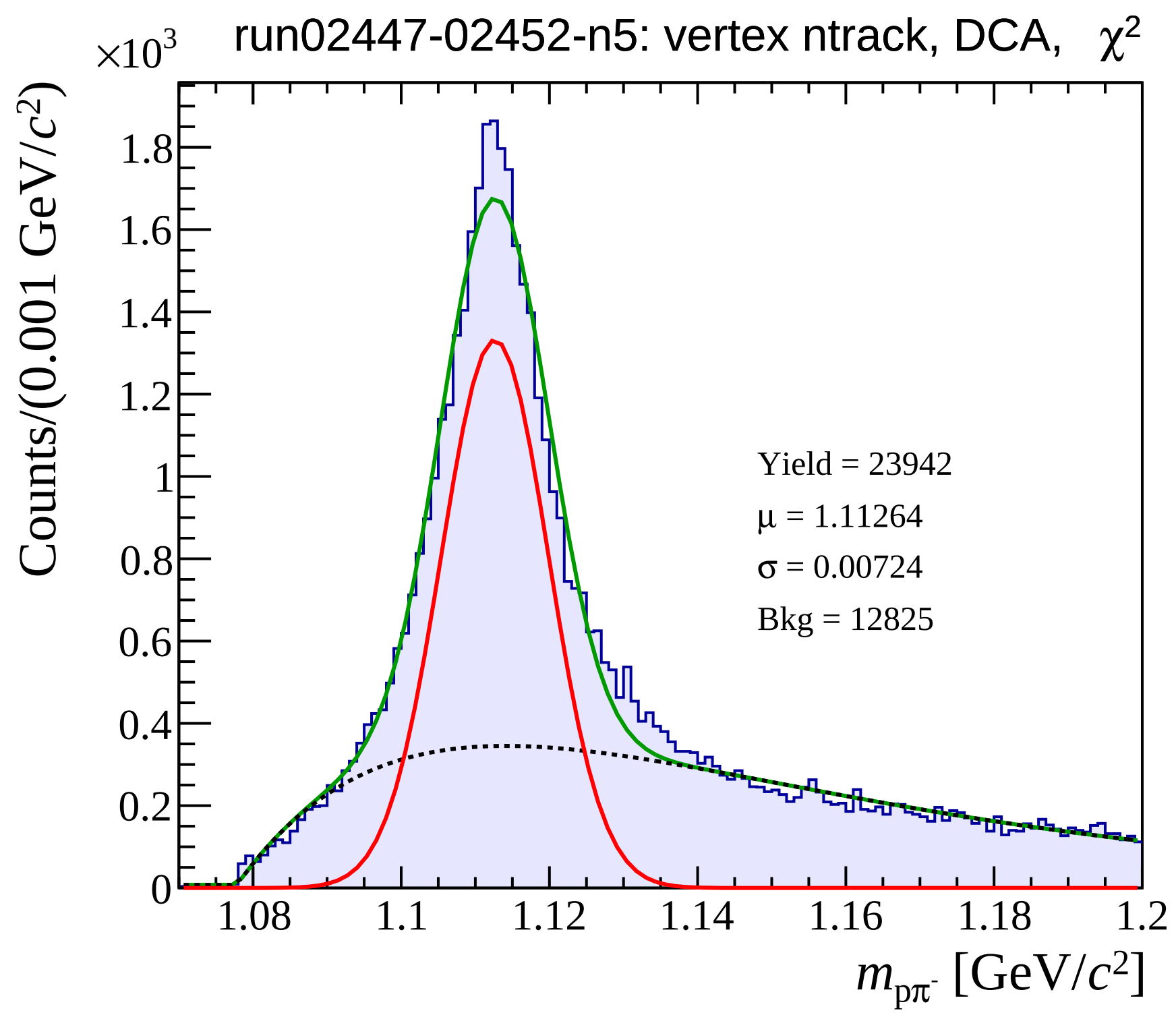

run02447-02452-n5: DCA, χ^2, MM ≥ 0.5
Counts/(0.001 GeV/c^2)
900
800
700
600
500
400
300
200
100
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 10623
μ = 1.11221
σ = 0.00559
Bkg = 2488

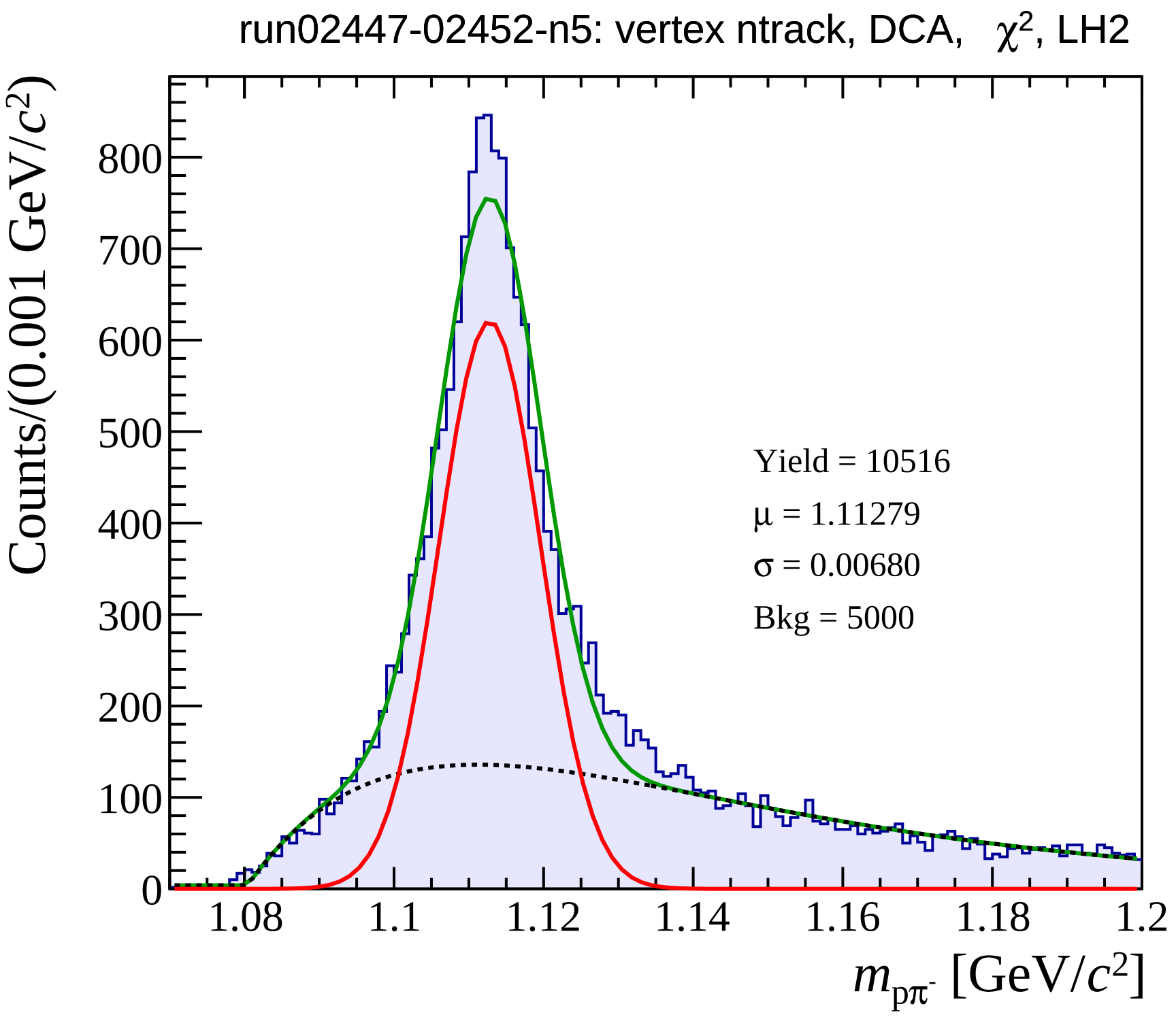

run02447-02452-n5: DCA, χ^2, LH2, MM ≥ 0.5
Counts/(0.001 GeV/c^2)
450
400
350
300
250
200
150
100
50
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 5159
μ = 1.11225
σ = 0.00541
Bkg = 770

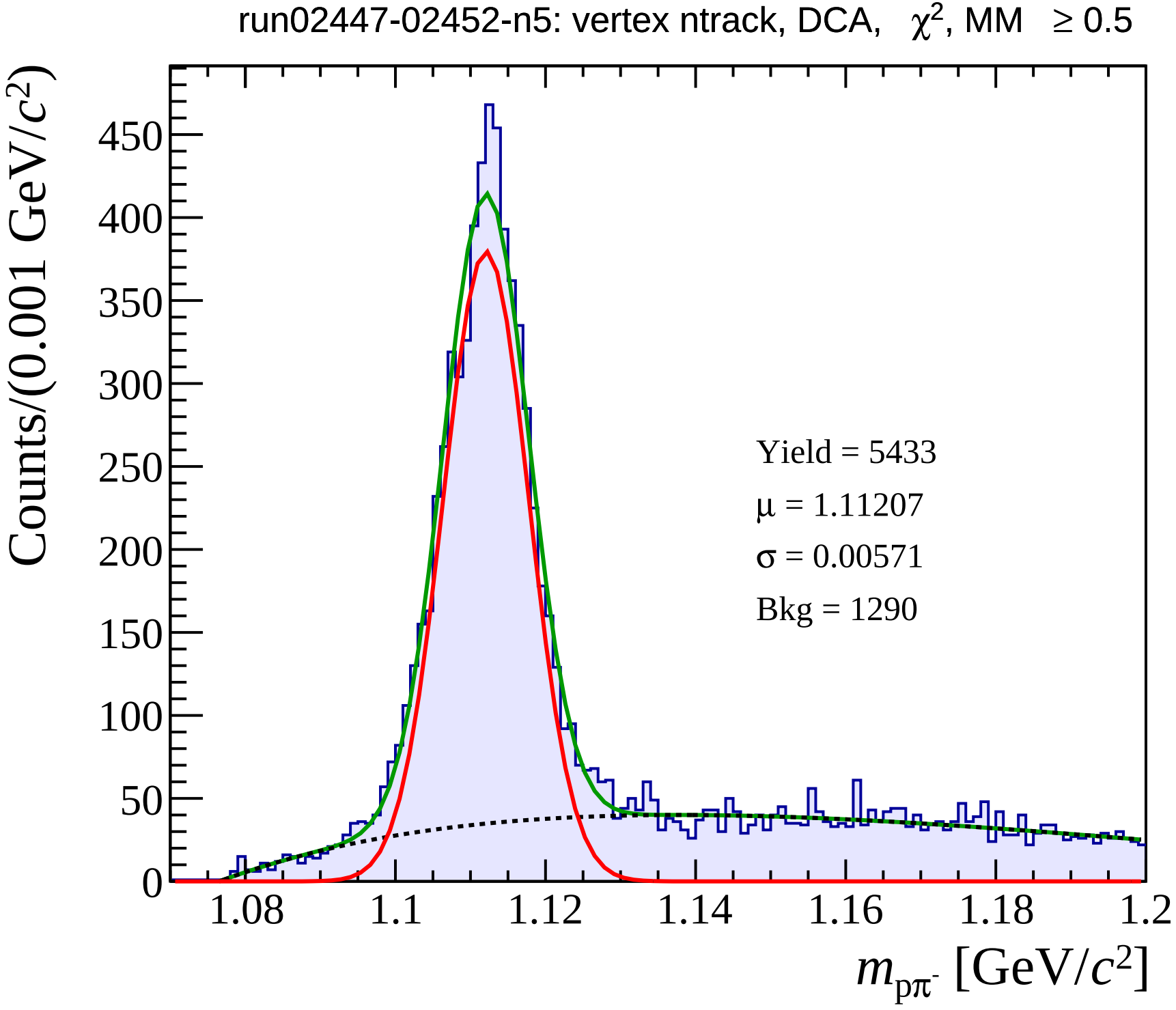

run02447-02452-n5: ntrack, DCA, χ^2, MM ≥ 0.5
Counts/(0.001 GeV/c^2)
600
500
400
300
200
100
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 6931
μ = 1.11205
σ = 0.00565
Bkg = 1518

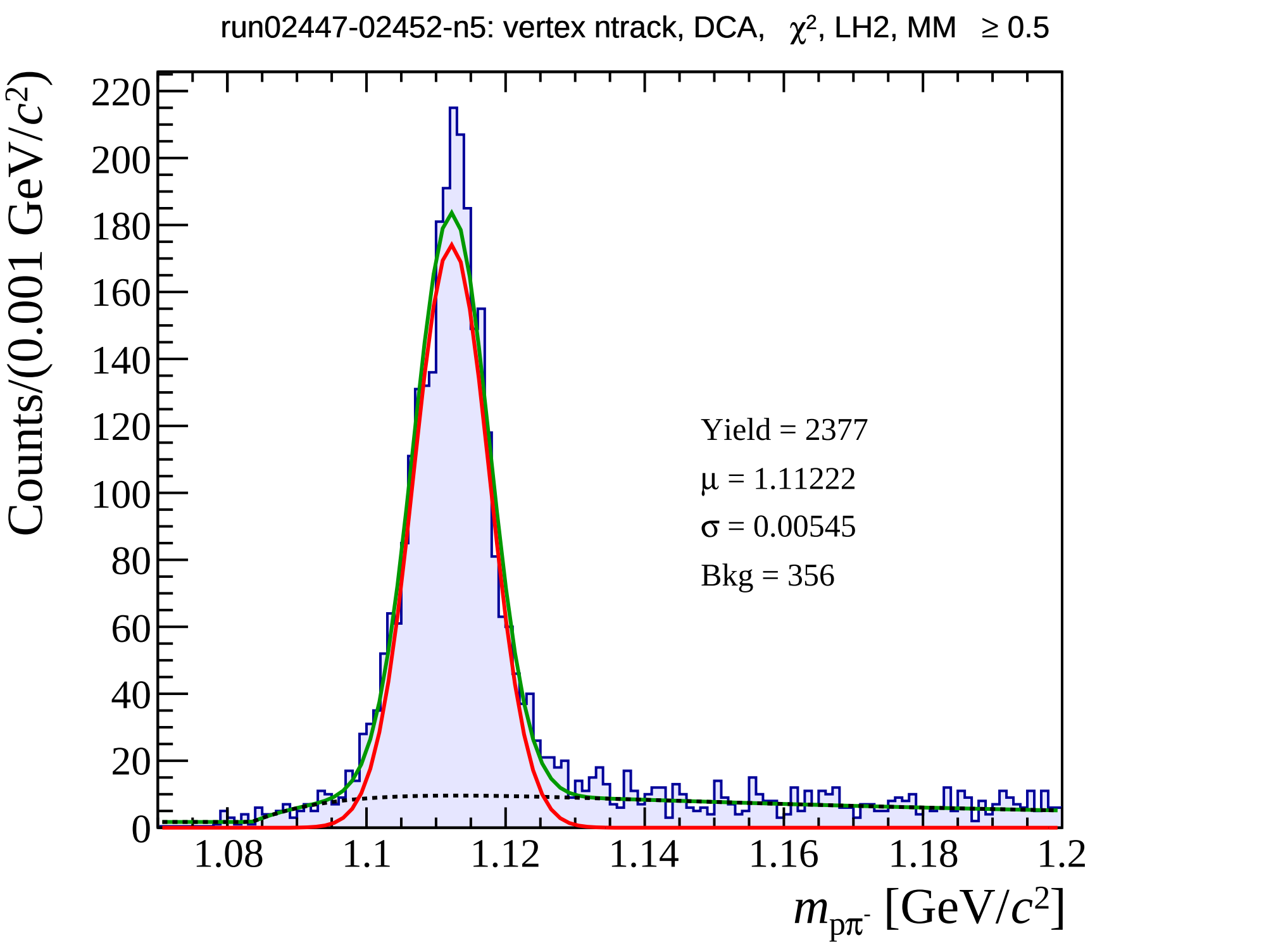

run02447-02452-n5: ntrack, DCA, χ^2, LH2, MM ≥ 0.5
Counts/(0.001 GeV/c^2)
300
250
200
150
100
50
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 3198
μ = 1.11210
σ = 0.00518
Bkg = 544

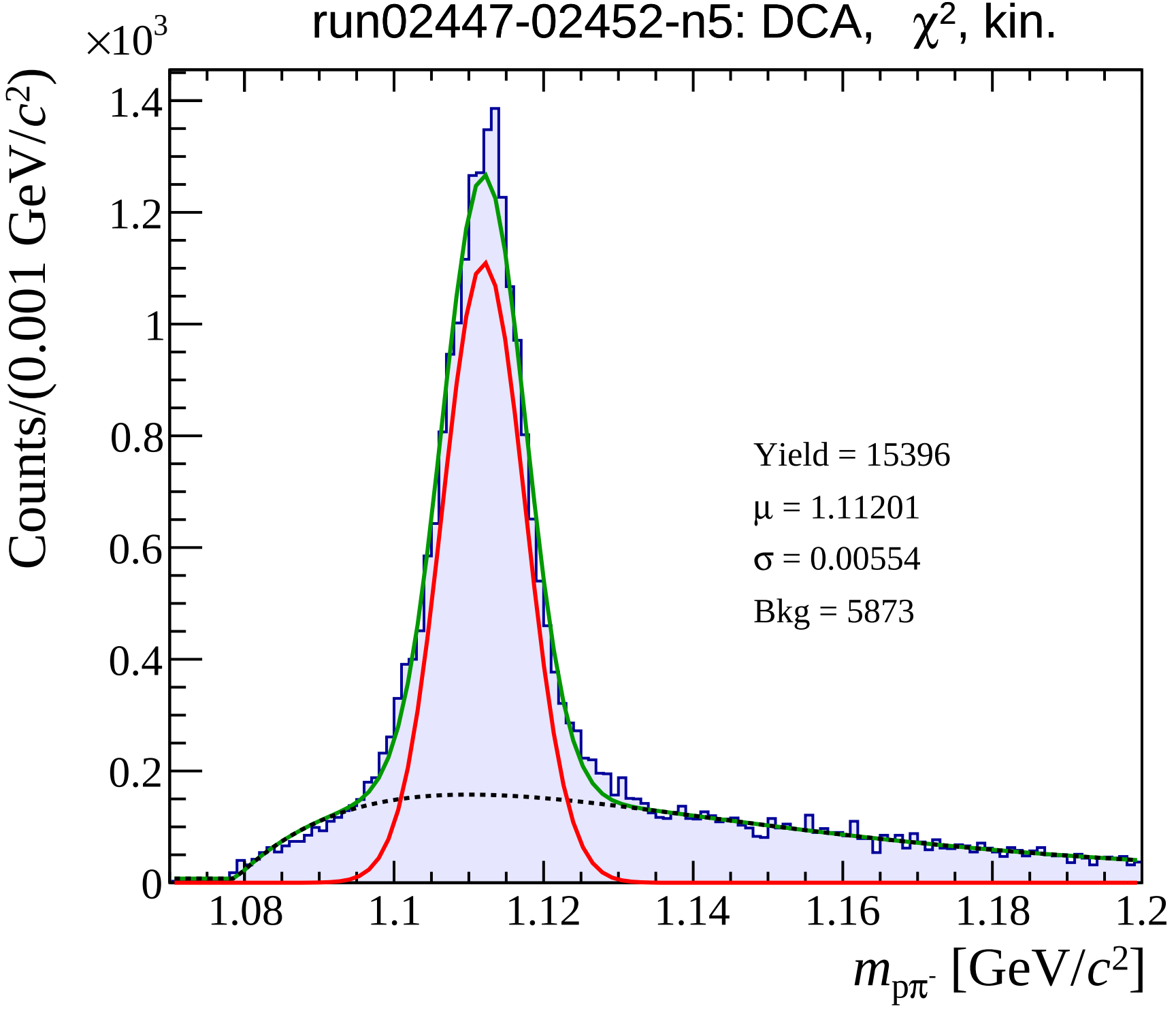

run02447-02452-n5: vertex ntrack
×10^3
Counts/(0.001 GeV/c²)
2.2
2
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m_pπ- [GeV/c²]
Yield = 27194
μ = 1.11223
σ = 0.00732
Bkg = 23051

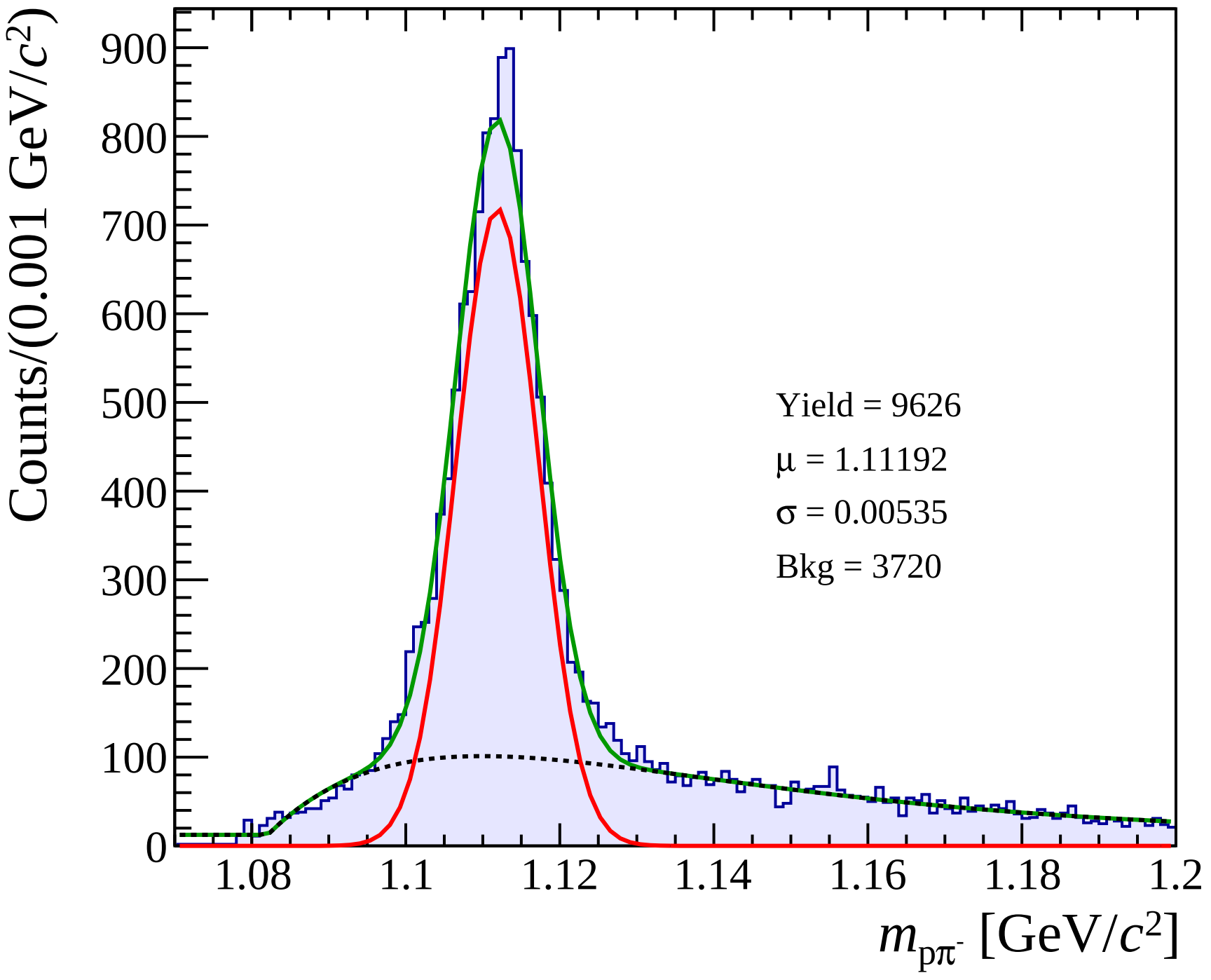

run02447-02452-n5: vertex ntrack, DCA, χ^2
$\times 10^3$
Counts/(0.001 GeV/c^2)
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 23942
μ = 1.11264
σ = 0.00724
Bkg = 12825

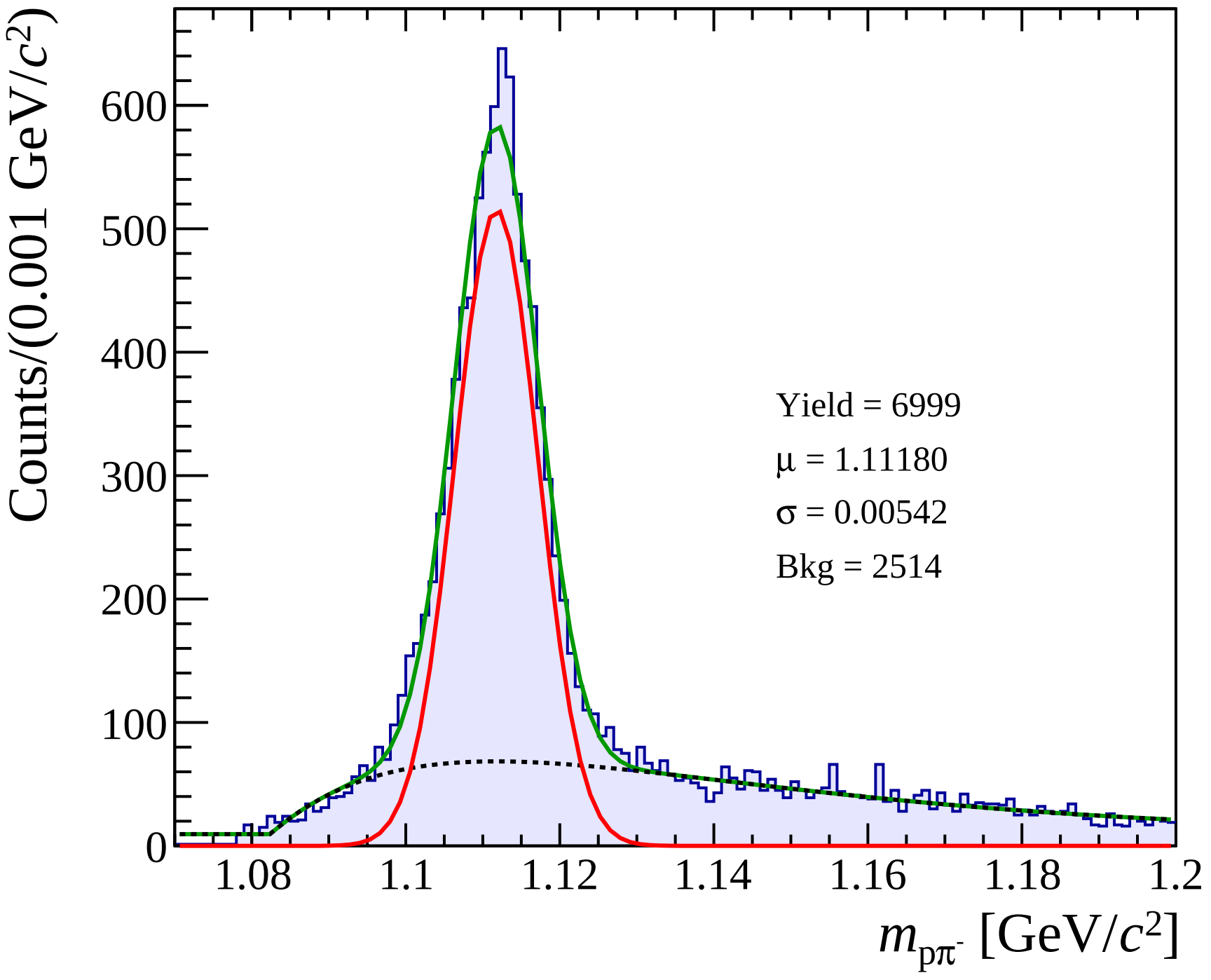

run02447-02452-n5: vertex ntrack, DCA, χ^2, LH2
Counts/(0.001 GeV/c^2)
0
100
200
300
400
500
600
700
800
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 10516
μ = 1.11279
σ = 0.00680
Bkg = 5000

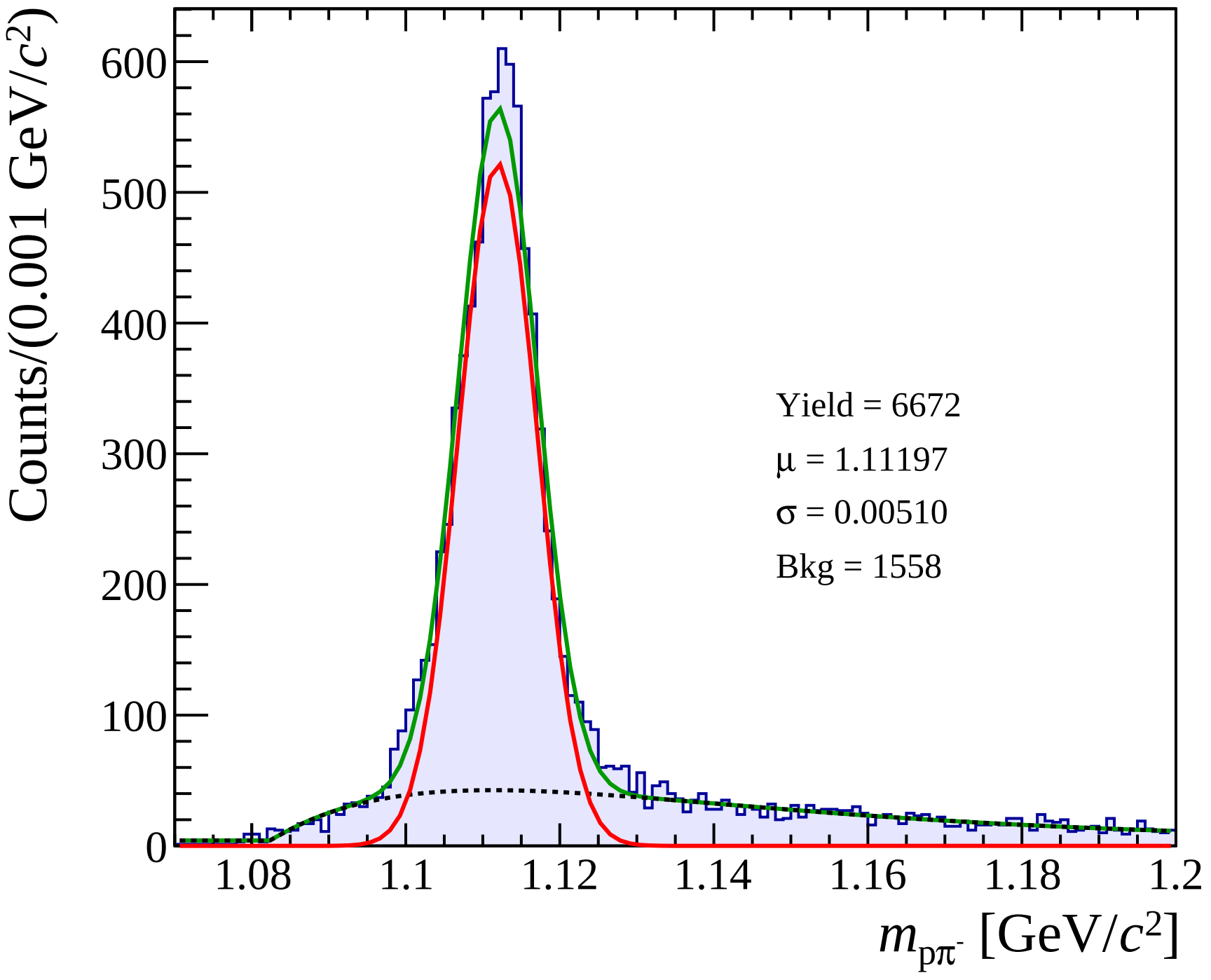

run02447-02452-n5: vertex ntrack, DCA, χ^2, MM ≥ 0.5
Counts/(0.001 GeV/c^2)
450
400
350
300
250
200
150
100
50
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 5433
μ = 1.11207
σ = 0.00571
Bkg = 1290

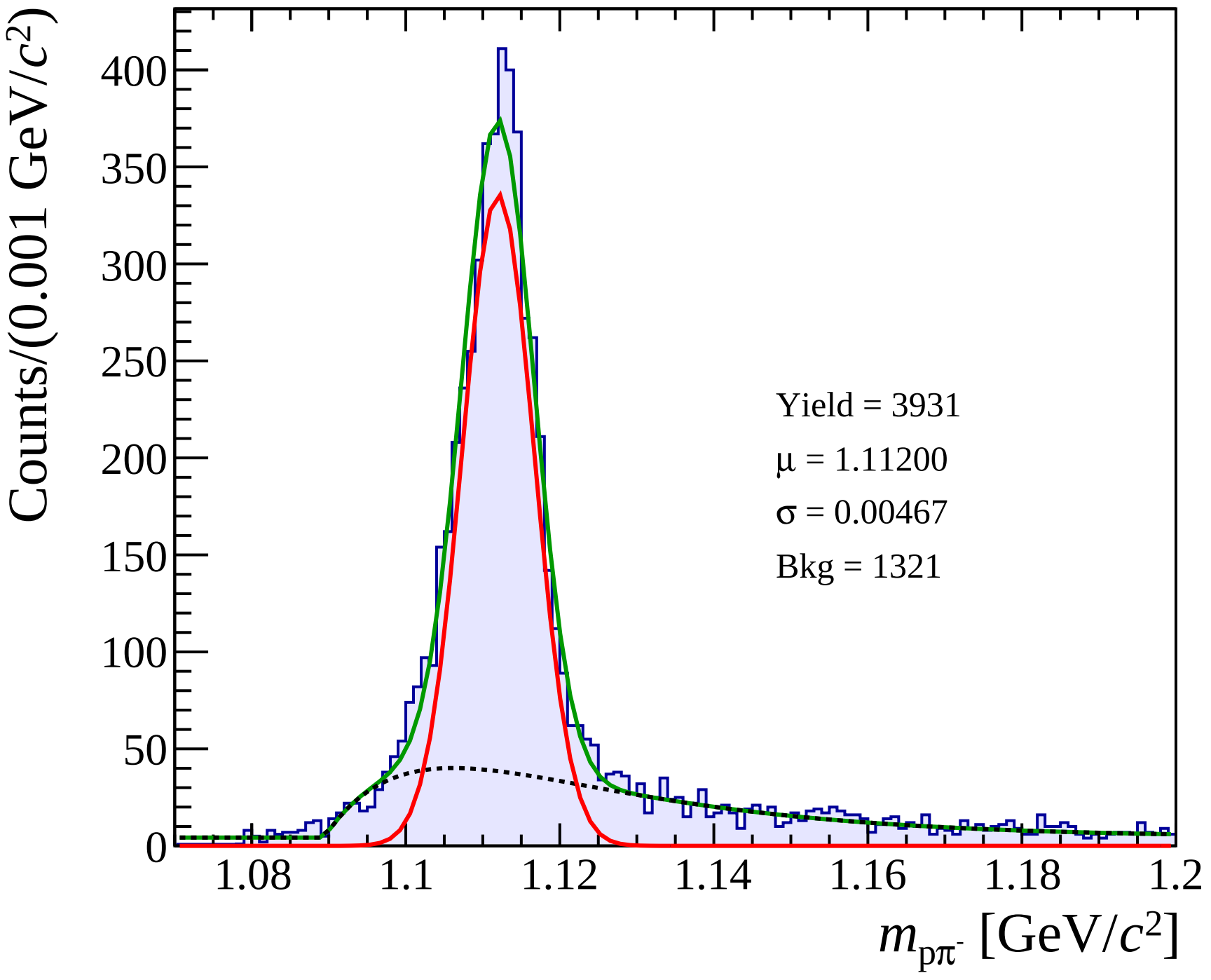

run02447-02452-n5: vertex ntrack, DCA, χ², LH2, MM ≥ 0.5
Counts/(0.001 GeV/c²)
220
200
180
160
140
120
100
80
60
40
20
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m pπ- [GeV/c²]
Yield = 2377
μ = 1.11222
σ = 0.00545
Bkg = 356

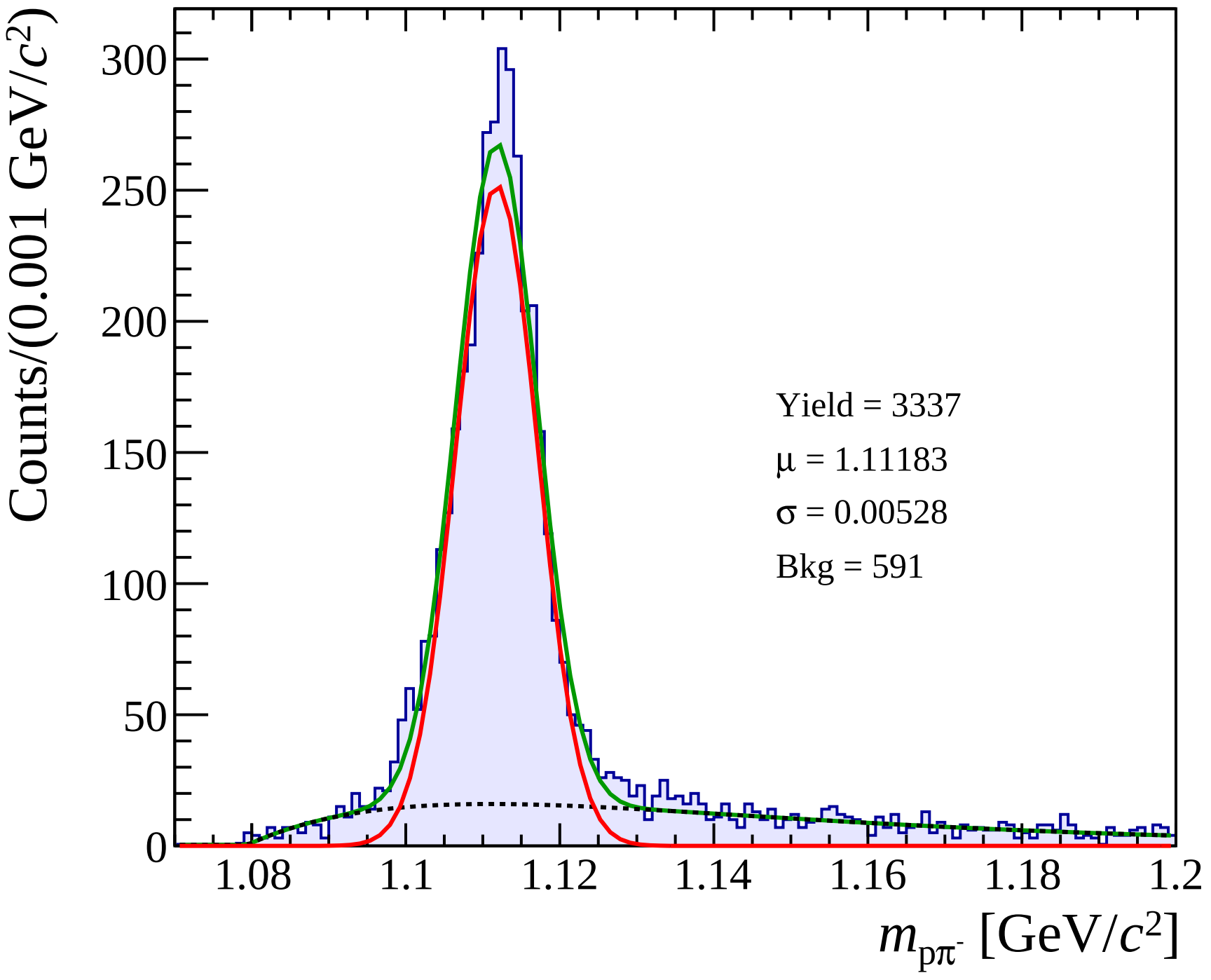

run02447-02452-n5: DCA, χ², kin.
×10³
Counts/(0.001 GeV/c²)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m pπ- [GeV/c²]
Yield = 15396
μ = 1.11201
σ = 0.00554
Bkg = 5873

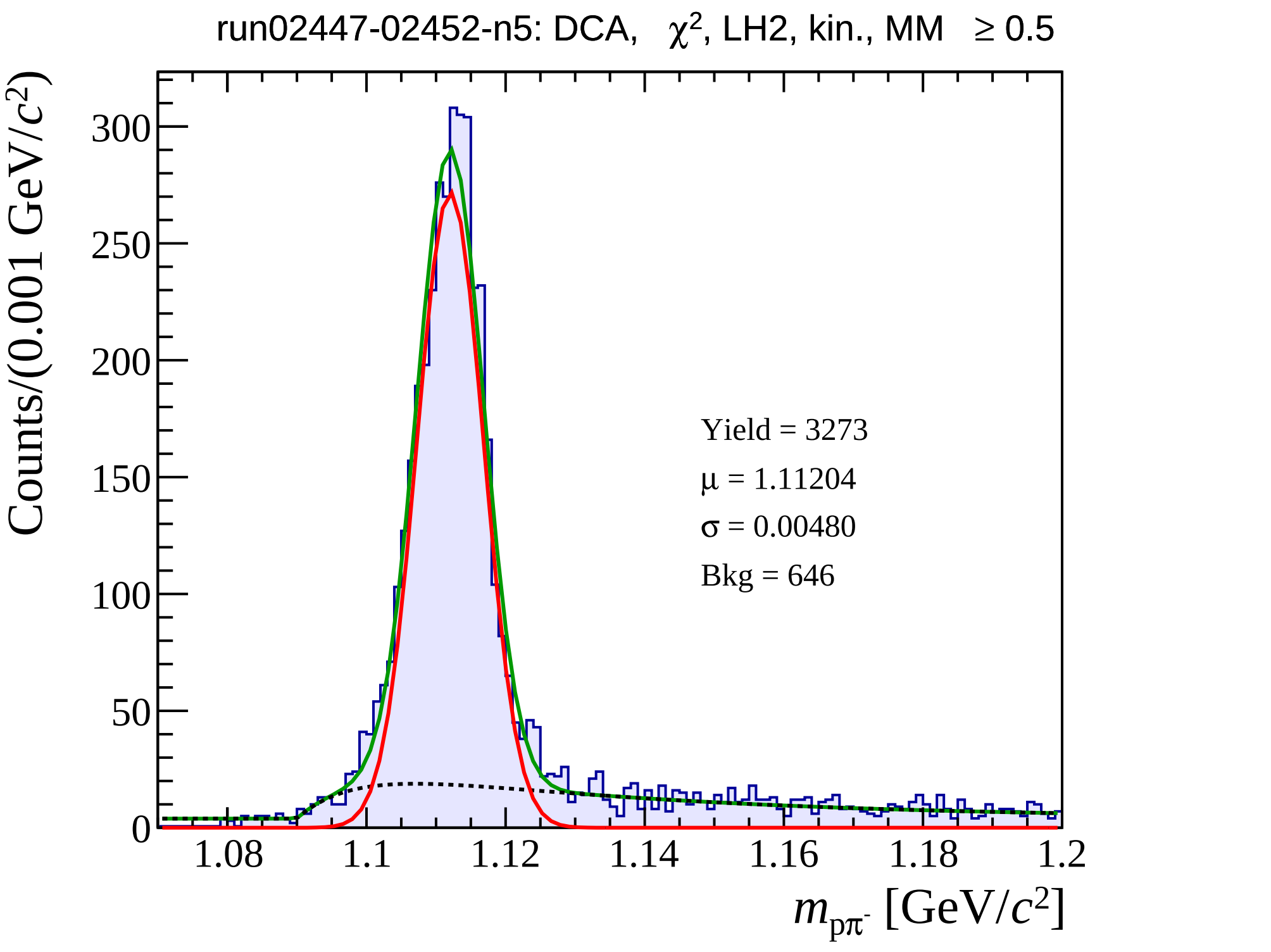

run02447-02452-n5: ntrack, DCA, χ², kin.
Counts/(0.001 GeV/c²)
900
800
700
600
500
400
300
200
100
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m pπ- [GeV/c²]
Yield = 9626
μ = 1.11192
σ = 0.00535
Bkg = 3720

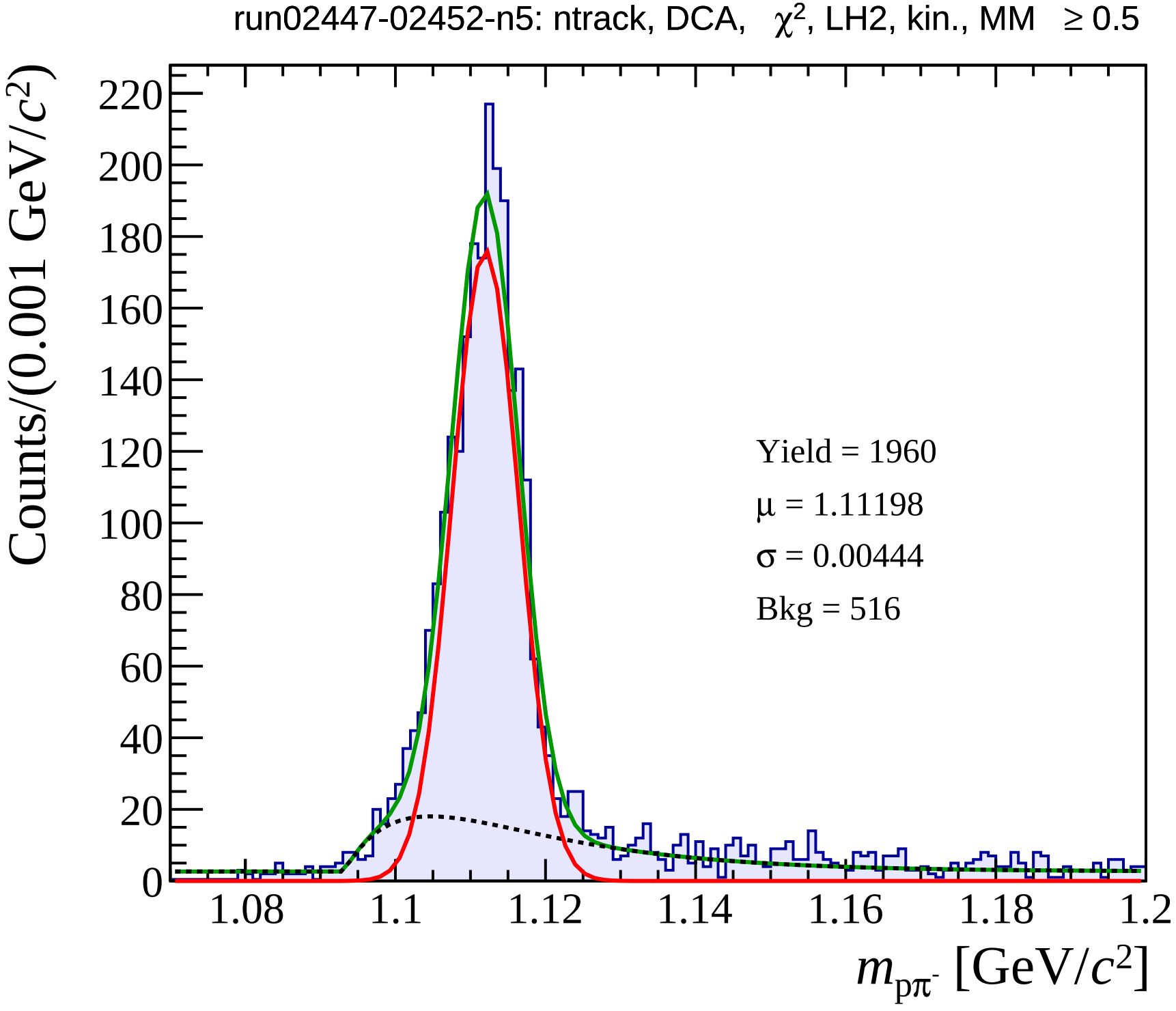

run02447-02452-n5: vertex ntrack, DCA, χ^2, kin.
Counts/(0.001 GeV/c^2)
600
500
400
300
200
100
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 6999
μ = 1.11180
σ = 0.00542
Bkg = 2514

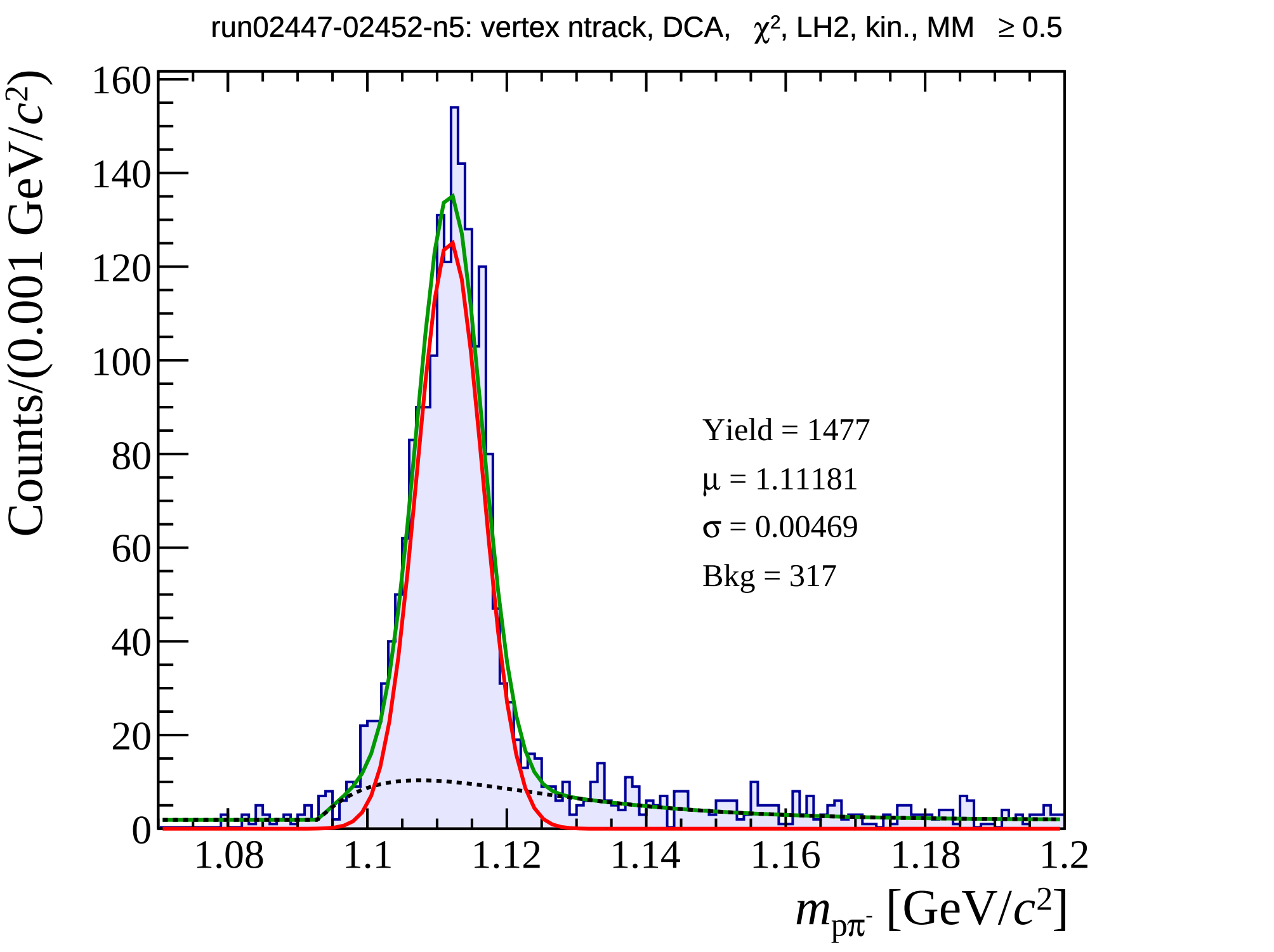

run02447-02452-n5: DCA, χ², LH2, kin.
Counts/(0.001 GeV/c²)
600
500
400
300
200
100
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m pπ- [GeV/c²]
Yield = 6672
μ = 1.11197
σ = 0.00510
Bkg = 1558


run02447-02452-n5: ntrack, DCA, χ², LH2, kin.
Counts/(0.001 GeV/c²)
400
350
300
250
200
150
100
50
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m pπ- [GeV/c²]
Yield = 3931
μ = 1.11200
σ = 0.00467
Bkg = 1321


run02447-02452-n5: vertex ntrack, DCA, χ^2, LH2, kin.
Counts/(0.001 GeV/c^2)
300
250
200
150
100
50
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 3337
μ = 1.11183
σ = 0.00528
Bkg = 591


run02447-02452-n5: DCA, χ^2, LH2, kin., MM ≥ 0.5
Counts/(0.001 GeV/c^2)
300
250
200
150
100
50
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 3273
μ = 1.11204
σ = 0.00480
Bkg = 646


run02447-02452-n5: ntrack, DCA, χ^2, LH2, kin., MM ≥ 0.5
Counts/(0.001 GeV/c^2)
220
200
180
160
140
120
100
80
60
40
20
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
$m_{p\pi^-}$ [GeV/c^2]
Yield = 1960
μ = 1.11198
σ = 0.00444
Bkg = 516


run02447-02452-n5: vertex ntrack, DCA, χ², LH2, kin., MM ≥ 0.5
Counts/(0.001 GeV/c²)
160
140
120
100
80
60
40
20
0
1.08
1.1
1.12
1.14
1.16
1.18
1.2
m pπ- [GeV/c²]
Yield = 1477
μ = 1.11181
σ = 0.00469
Bkg = 317